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Unveiling cybersecurity mysteries: A comprehensive survey on digital forensics trends, threats, and solutions in network security

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ABSTRACT

The field of digital forensics is undergoing a paradigm shift because security breaches are now occurring outside of conventional domains such as mobile devices, databases, networks, multimedia, cloud platforms, and the Internet of Things (IoT) all require a complete approach. This study report reveals a high level of ambiguities and process redundancies within the subdomains of digital forensics through the completion of a Systematic Literature Review (SLR). To address this, we suggest a high-level theoretical metamodel that unifies tasks, operations, procedures, and methods of research across many subdomains that will help forensic investigators during digital investigations to organize and integrate evidence. The study also discusses the necessity of global perspectives in research on digital forensics and provides a qualitative evaluation of past surveys, highlighting similar difficulties, obstacles, and key issues across domains, whereas earlier surveys concentrated on domains. The findings through examination offer a multidimensional knowledge of the difficulties in digital forensics and suggested metamodels help to create a more cohesive and integrated approach to digital investigations, establishing an environment for further study and collaborations in this crucial domain.

1. Introduction

Over the last few years, the rapid growth in cybercrime as well as increasing the appositeness of digital forensics have become increasingly in demand in recent years because of the explosive rise in cybercrime and the growing applicability of digital devices to traditional criminal investigation (Casino and Dasaklis, 2022; Laksmi et al., 2025). Digital forensics (DF), is a specialized field focused on identifying digital evidence and devices storing data digitally, using validated processes for legal applications both within and outside of a courtroom (Hafiz Deandra and Muda, 2025; Singh et al., 2019; Tiwari et al., 2021). According to the International Telecommunication Union (ITU, 2024), global internet usage reached 5.4 billion people in 2023, accounting for approximately 67 % of the world's population. Complementing this, Cisco's latest "Annual Internet Report" shows that at the end of 2023, the total network-connected devices will be 29.3 billion (Cisco and Cisco, 2020), and the norm of traffic flow controlled by the system will be almost 50 GB per month (Cisco, 2021).

In this multifaceted security landscape, digital forensics plays a crucial role in securing, protecting, and monitoring network investigations (Nour and Pourzandi, 2023), aiding investigators in navigating high-volume data to uncover pertinent evidence (Wasim Malik et al., 2024). Digital evidence can emanate from various devices electronically connected to a network, including those under examination and those in smart environments (Kebande and Surveys, 2024; Pandey et al., 2024). Filesystem directories offer access to low-level data, enabling investigators to trace machine activity and interpret how and where a device has been used previously. Additionally, smart environments may hold collaborative data associated with multiple devices and evidence not locally saved on the machine (Du et al., 2020; Stoyanova et al., 2020). Network security is becoming an essential challenge in the modern period, which has led to enterprises adopting it widely (Hosseini, 2019).

As organizations increasingly rely on interconnected systems, the potential for security breaches grows. To address this, network security tools such as Wireshark, Snort, Tcpdump, and Capsa play a pivotal role in capturing and analyzing digital evidence. These tools provide critical insights into cyber-attacks, aiding investigators in reconstructing the digital transmission chain and understanding attacker methodologies (Mishra et al., 2025). By leveraging such tools, organizations can fortify their defenses against evolving threats. To secure the data network, digital forensics tools not only record illicit activities (Chen, 2019) but

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also facilitate the comprehensive investigation of the tactics, techniques, and procedures employed by cybercriminals. This process is instrumental in the identification of individuals involved in the attacks.

In the current landscape, as cybercrime continues to rise, digital evidence becomes a cornerstone at crime scenes. Various digital devices, such as email platforms, e-money services, flash drives, cloud service providers, and electronic devices, frequently yield crucial digital evidence (Casino and Dasaklis, 2022; Laksmi et al., 2025). The seamless integration of network and cyber security with digital forensics is imperative to secure data effectively (Arif et al., 2025). By doing so, organizations can better protect themselves from evolving cyber threats and ensure the integrity of digital evidence, which serves as key proof in investigating and prosecuting various offenses (Malatras et al., 2023).

In summary, this study highlights the crucial role of digital forensics in addressing modern cyber threats. By leveraging cutting-edge tools and interdisciplinary methods, it enables the detection and resolution of cybercrimes across diverse domains. This survey offers a concise yet structured overview, bridging the gap between technical progress and practical application. Such clarity and integration are essential for guiding researchers and practitioners in navigating the complexities of this rapidly evolving field.

1.1. Motivation

Digital evidence has become a cornerstone of contemporary crime investigations, and the varied nature of digital proof requires diverse strategies and techniques for analysis and processing. Collecting digital evidence from different domains has prompted extensive research, addressing challenges specific to each domain. The motivation behind this study is to present a comprehensive survey of digital forensics in each domain, offering insights into various aspects and providing directions for future research in this rapidly evolving field. The significance of this study lies in its emphasis on the pivotal roles digital forensics plays in ensuring the integrity of personal data and protecting it from data breaches and unknown attackers. The study explores the distinct challenges faced in different domains, aiming to guide future researchers by examining state-of-the-art tools, addressing current issues, and outlining future directions in digital forensics.

As the prevalence of cyber threats continues to escalate, the need for robust digital forensics practices becomes more pronounced. Understanding the nuances of digital evidence collection and analysis in diverse domains is crucial for staying ahead of evolving cyber threats. The multifaceted nature of digital forensics, spanning network security, smart environments, and various digital devices, highlights its broad applicability and underscores its role in securing digital information. By delving into these domains and addressing their specific challenges, researchers can contribute to the ongoing development of effective digital forensics methodologies, ensuring the continued integrity and security of digital data.

1.2. Research focus

This study seeks to analyze the current state of digital forensics through a detailed survey of its core subdomains, structured around key questions that aim to systematically explore this evolving field. The research is guided by the following key questions:

1. What are the primary challenges associated with digital forensics across different subdomains?
2. How do existing tools, frameworks, and methodologies address these challenges, and what are their limitations?
3. What research gaps exist in digital forensics, and what directions can future studies explore to enhance the field's efficacy and relevance?

To address these questions, this study focuses on:

- Identifying challenges and complexities across major digital forensics subdomains, including network, mobile, database, IoT, cloud, and multimedia forensics.
- Evaluating existing tools, frameworks, and methodologies to uncover their strengths, limitations, and applicability in real-world scenario.
- Exploring opportunities for innovation by addressing research gaps and proposing research directions and potential solutions for future studies.

These focal points aim to provide a structured understanding of the evolving landscape of digital forensics while fostering advancements in the field. The study also underscores the global relevance of establishing standardized frameworks to address jurisdictional complexities and ensure cross-border applicability of digital forensic techniques.

1.3. Selection criteria

To ensure a comprehensive and representative survey, the following systematic approach was adopted:

1. **Databases Searched:** Comprehensive searches were conducted across prominent academic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Scopus. These platforms were chosen for their extensive coverage of high-impact publications in digital forensics and cybersecurity.
2. **Search Keywords:** Keywords included "digital forensics," "cybersecurity," "network forensics," "IoT forensics," "cloud forensics," and for specific methodologies such as "artificial intelligence in forensics", "blockchain in digital investigations." Boolean operators (e.g., AND, OR) and database-specific filters were used to refine results.
3. **Inclusion Criteria:** Papers published between 2019 and 2025, focusing on the intersection of digital forensics and cybersecurity, with empirical validation, novel frameworks, or critical reviews. Only peer-reviewed journal articles, conference proceedings, and book chapters were considered.
4. **Exclusion Criteria:** Outdated or irrelevant papers, non-English publications, and studies lacking empirical evidence or practical applications were excluded.
5. **Screening and Assessment:** Titles and abstracts were initially screened to eliminate irrelevant works, followed by a full-text analysis of shortlisted papers. Reference lists of key studies were reviewed to capture additional significant works.
6. **Diversity and Coverage:** Selected works spanned multiple subdomains, including IoT, cloud, and multimedia forensics, ensuring a holistic perspective and a balance of theoretical advancements and practical implementations.

Fig. 1. Hierarchical representation of key concepts in the systematic paper selection process.

The systematic selection process ensures the survey's depth, breadth, and relevance to the evolving field of digital forensics, as depicted in Fig. 1. The visual representation clarifies the methodology by highlighting the interconnected steps, from database searches to the inclusion of diverse and impactful works.

1.4. Paper organization

This study is organized to systematically address the multifaceted domain of digital forensics. Section 2 provides the background, offering an in-depth overview of digital forensics, its subdomains, and its critical role in cybersecurity. Section 3 presents the related work, summarizing key contributions, methodologies, and research efforts in the field, setting the stage for subsequent analysis. Section 4 explores recent tools, techniques, and methodologies, showcasing their applications across various forensic domains such as network, mobile, database, IoT, cloud, and multimedia forensics. Section 5 delves into discussion, challenges, and future directions, addressing existing gaps, emerging trends, and actionable recommendations for advancing the field. Finally, Section 6 concludes the paper by summarizing the key findings, implications, and contributions, providing a roadmap for future research and practice in digital forensics.

2. Background

Digital forensics is a rapidly growing field essential to combating cyber-attacks in today's interconnected world. With increasing reliance on digital systems, digital forensics help to secure data and digital infrastructures from malicious actors. However, the field also faces evolving challenges in evidence collection, investigation, and data preservation. Addressing these challenges requires continuous development of tools and methodologies suited for modern digital ecosystems. For a comprehensive overview of foundational concepts in digital forensics, readers are referred to recent survey (Rafifah Syaakirah et al., 2025), which offers in-depth discussion of advancements and challenges. Thus, in this background study, we will provide an overview of DF along with various aspects of digital forensics and talk about the challenges and security issues as shown in Fig. 2.

2.1. Digital forensics

Digital Forensics or computer forensics is a branch of forensic science and cyber security that focuses on the recovery of various applications of investigation of mate-rials or preserving evidence found in digital or other electronic devices related to cybercrimes (Choi and Sciences, 2022; Javed et al., 2022). Previously, the term digital forensic was mainly used as computer forensics that focused on performing a structured investigation after analyzing and maintaining a documented chain of evidence to find out exactly when and what happened on a computer and who did it. But now it has expanded to cover the investigation of all electronic devices that store data digitally (Kebande, 2022; Kebande et al., 2022). Digital forensics is a process of investigation to protect your data from being stolen digitally or electronically (Rizvi et al., 2022). As the crime ratio increases daily, digital forensics helped in search of any unusual activity that caused crimes like sexual assault, stolen data, and civil and criminal cases (Baror et al., 2021; 2021). There are various devices where digital forensics play a vital role in keeping your documentation preserved, protected, and easy to investigate.

Digital forensics is also widely used in legal entity investigations such as civil and criminal investigation processes as it becomes an important aspect of law enforcement agencies and businesses as society increases its dependency on computer systems or cloud computing (Hemdan and Manjaiah, 2021; Ramos Brandao, 2019). Imagine if security violence happens at a company or an organization, causing a data spill. In these circumstances, a digital forensics analyst would come in and discover how attackers obtained access to the network, where they came over to the data, and what they did to the network, whether they took any information or deep-rooted malware. Under those conditions, a digital forensic analyst runs to earth to recover the data documents, private files, images, emails, and other important data from computer hard discs and other data drive devices (such as compressed files, thumb drives), with deleted, destructed, spoiled, missed or other diversely manipulated evidence.

Furthermore, Digital forensics is all about gathering the relevant

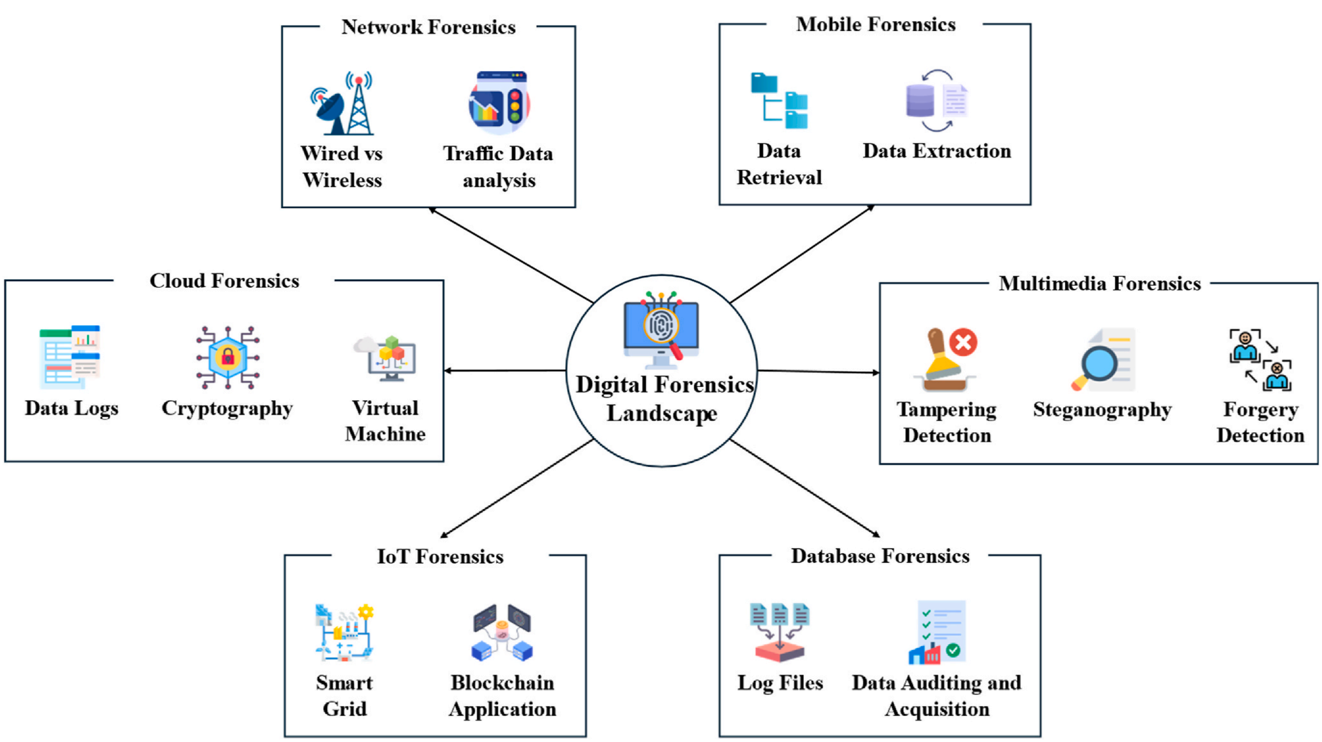

Fig. 2. The landscape of digital forensic: Diverse aspects mapping cyber threats.

information that maintains its integrity then identifying, preserving, examining, and analyzing the data and documenting digital evidence essentially with legal compliance guidelines to make information admissible to be validated in the process to present in a court of law when required (Rana et al., 2023). Besides, digitally connected devices store massive amounts of data such as log devices which usually perform all actions by their users as well as autonomous activities performed by devices such as network connection and data transfers including vehicles, android devices, routers, computers, traffic signals and many other devices throughout the public and private circle (de Azambuja et al., 2023). Digital forensics has a much larger impact on both the digital and physical world.

Consequently, the rapid evolution of security incidents across digital subdomains such as Mobile, Network, Filesystem, Memory, Database, Internet-of-Things, Cloud-based environments, AI, Blockchain, Multimedia, and across-the-edge network designs has created an urgent need for advanced DF models, tools, and techniques to retrieve diverse artifacts and evidence (Stoyanova et al., 2020). While prior studies have individually addressed specific subdomains or aspects of digital forensics, a holistic framework encompassing the field in its entirety remains absent. Thus, building on the foundational works summarized in Table 1, which highlights the key contributions and focus areas within the digital forensics landscape, this paper provides a structured and comprehensive exploration of the field. It seeks to bridge that gap with the following contributions:

- We conduct a comprehensive review of six core areas of digital forensics: network, mobile, database, IoT, cloud, and multimedia forensics. For each subdomain, we discuss its unique challenges, tools, methodologies, and emerging trends.
- The paper identifies common challenges and solutions across these areas, highlighting overlaps, interdependencies, and unique characteristics of subdomains. This integrated approach fosters a holistic understanding of digital forensics as a multidisciplinary field.
- We propose a digital forensics workflow framework that outlines the five main phases of investigation: identification, preservation, analysis, documentation, and presentation. This framework acts as a practical guide and highlights critical points where challenges and redundancies often arise.
- By analyzing existing research, we uncover gaps that require attention, particularly in newer fields such as IoT and multimedia forensics. Additionally, the paper suggests potential directions for addressing these gaps through innovation and interdisciplinary collaboration.
- In addition to theoretical discussions, the paper emphasizes the applicability of its findings to practical challenges faced by forensic investigators. This ensures that the insights and recommendations are directly relevant to both academic research and real-world applications.

Unlike previous surveys that primarily focus on individual forensic subdomains or limited technical perspectives, this paper offers a unified and cross-domain analysis that integrates theoretical frameworks, domain-specific challenges, and practical investigative workflows. By emphasizing standardization gaps, missing investigative phases, and underexplored domains like multimedia and IoT forensics, the study sets itself apart with a structured metamodel and future-ready roadmap tailored for both research and field application. This multidimensional focus enables a deeper and more actionable understanding of the digital forensics landscape. This paper ultimately consolidates diverse insights into a cohesive roadmap that addresses pressing challenges in digital forensics. Its multidisciplinary scope, spanning domains such as network, mobile, IoT, and multimedia forensics, reinforces the field's central role in safeguarding digital ecosystems. By confronting methodological gaps and emphasizing actionable workflows, this work equips both researchers and practitioners to navigate an increasingly complex cybersecurity landscape. Thus, this study distinguishes itself by bridging fragmented subdomain research through an integrated forensic metamodel that transforms isolated findings into a cohesive framework rarely seen in existing literature.

Table 1
Highlights key contributions from foundational studies, illustrating the scope and focus areas of digital forensics research.

Author(s)	Focus Area	Main Contribution	Limitations
Choi and Sciences (2022)	File Recovery in Damaged Systems	Proposed NTFS-based recovery for damaged RAID systems to preserve evidence.	Limited to RAID systems, lacks adaptability to diverse architectures.
Javed et al. (2022)	Evolution of Computer Forensics	Comprehensive survey on computer forensics' evolution, tools, and challenges.	Does not cover emerging fields like IoT and cloud forensics.
Kebande et al. (2022)	Cloud Forensic Readiness	Introduced a finite state model for cloud forensic readiness.	Overlook multi-cloud environments and jurisdictional complexities.
Kebande (2022)	Industrial IoT Forensics	Developed forensic techniques for Industry 4.0 IoT systems.	Limited to industrial IoT; lacks focus on consumer IoT issues.
Rizvi et al. (2022)	AI Applications in Network Forensics	Explored AI-driven approaches for analyzing network forensic activities.	Does not fully explore anti-forensic methods used by attackers.
Baror et al. (2021)	Digital Evidence in Smart Cities	Proposed digital evidence frameworks for diverse environments.	Scalability challenges in large, distributed environments.
(Baror et al., 2021)	Crowd-Sourced Digital Evidence	Presented a traceable model for digital crowd-sourced forensics.	Challenges in maintaining traceability and evidence integrity.
Ramos Brandao (2019)	Cloud Forensics Challenges	Highlighted technical barriers to cloud forensic investigations.	Lacks practical solutions for cross-border evidence handling.
Hemdan and Manjaiah (2021)	Digital Forensic Models in Cloud	Proposed efficient models for investigating cybercrimes in cloud computing.	Limited focus on hybrid clouds and large-scale scenarios.
Rana et al. (2023)	Blockchain for Digital Evidence	Proposed blockchain-based secure evidence storage using smart contracts.	Limited by blockchain scalability and energy efficiency issues.
de Azambuja et al. (2023)	AI in Cybersecurity	Surveyed AI anomaly detection techniques for Industry 4.0 cybersecurity.	Overlooks privacy-preserving AI techniques for forensic applications.
(A Al-Dhaqm et al., 2021)	Subdomains of Digital Forensics	Reviewed forensic tools, gaps, and future directions for the field.	Lacks case studies for IoT and multimedia forensics.

2.2. Core subdomains of digital forensics

2.2.1. Network forensics

Network forensics is a cornerstone of digital forensics, focusing on analyzing network-based evidence to identify and mitigate security breaches. It deals with all kinds of network-based inspections, including analyzation, abstraction, explication, case regeneration, investigation, and authentication, ensuring the evidential value and wholeness of compiled data. This enables the authentication and interpretation of network incidents with a high exploratory evidential load of prospective network irregularities (Lagrasse and Singh, 2020; Munkhondya, 2020; Qureshi and Tunio, 2021). As organizations increasingly rely on interconnected systems, network forensics has become essential for investigating cybercrimes and ensuring the security of digital infrastructures.

2.2.2. Mobile forensics

With the growing reliance on mobile devices in daily life, mobile forensics has become a critical subdomain of digital forensics, addressing crimes that involve portable devices. It involves the retrieval and analysis of data or evidence from devices with communication and internal memory capabilities, such as smartphones, tablets, GPS, and PDA devices. These devices often hold key evidence in criminal investigations, particularly with the increasing use of portable computing devices in crimes over the past years (A Al-Dhaqm et al., 2020b; Al-Dhaqm and Razak, 2020a Dinnur Rahman and Riadi, 2019). By enabling investigators to extract and preserve vital evidence, mobile forensics play a crucial role in modern investigations and combat the growing sophistication of mobile-based crimes.

2.2.3. Database forensics

As data breaches and fraud often target databases, database forensics has emerged as a pivotal field within digital forensics to uncover and analyze evidence stored in structured systems. It plays a pivotal role in uncovering fraudulent activities within databases, and extensive research has been dedicated to developing theories, structures, procedures, and techniques for effective database investigations. The multidimensional and complex nature of Database Management Systems (DBMS) presents unique challenges, necessitating a harmonized model for database forensics (A Al-Dhaqm et al., 2020a; 2020cAl-Dhaqm and Razak, 2020; Arafat Al-Dhaqm et al., 2020). By addressing these challenges, database forensics ensures the integrity of stored data and supports efforts to trace and prevent fraudulent activities.

2.2.4. IoT forensics

As IoT devices become widespread in personal and professional environments, IoT forensics has become indispensable for collecting and analyzing evidence from these interconnected systems. Unlike traditional digital forensics, IoT forensics encompasses a broader spectrum of evidence sources, including traffic signals, medical implants, smart home appliances, monitoring systems, and various IoT situations (A Akinbi, 2020; Stoyanova et al., 2020). This investigation delves into hardware redundancy, such as sensors, networked cars, RFIDs, and drones (J Hou, 2019). Despite evident challenges, current digital forensics tools and methodologies have not fully integrated IoT forensics, due to the scattered and heterogeneous nature of IoT infrastructures, creating difficulties for law enforcement agencies and investigators in gathering, searching, and analyzing evidence in IoT contexts (Akinbi and MacDermott, 2022; Alenezi et al., 2019; Dawson and A Akinbi, 2021; Sengan et al., 2023). Standardizing IoT forensic methodologies will be crucial for addressing the unique challenges posed by heterogeneous and distributed IoT systems.

2.2.5. Cloud forensics

Cloud forensics addresses the challenges of collecting and analyzing evidence stored in remote and distributed environments, particularly as cloud computing becomes ubiquitous. It is a multifaceted field that delves into criminal investigations and civil disputes involving crimes that leverage cloud resources, such as storing records of terrorist attacks. The physical location of hardware resources becomes intricate, and jurisdictional restrictions add more layers of difficulty, particularly when dealing with data stored in other countries (Manral et al., 2019). As cloud adoption continues to rise, robust forensic frameworks are essential to overcome jurisdictional and technical challenges in cloud investigations.

2.2.6. Multimedia forensics

Multimedia forensics plays a vital role in verifying the authenticity of digital media, addressing forgery, and ensuring the integrity of multimedia content. It is a specialized domain within digital forensics that focuses on examining and analyzing various multimedia substances. These efforts serve to identify potential risks, confirm authenticity, and detect tampering within multimedia files, as these can be altered or exploited for malicious purposes. The core area of focus in multimedia forensics is picture forgery detection, attracting extensive research interest (Bourouis et al., 2020; Costa et al., 2020; Saber et al., 2020). As digital media continues to spread, multimedia forensics is integral to maintaining trust and security in the digital ecosystem.

2.3. Digital forensics workflow

Digital forensics investigations can be broadly divided into five distinct phases, providing a structured approach to gathering, analyzing, and presenting evidence for legal and organizational purposes. This framework ensures a methodical process while highlighting the phases where potential threats, issues, and challenges may arise.

1. ***Identification:*** The identification phase involves recognizing potential sources of digital evidence, such as devices, logs, or network traffic, and determining their relevance to the investigation. This phase often faces challenges in heterogeneous environments like IoT systems and cloud platforms, where evidence sources can be highly distributed and dynamic. Threats during this stage include evidence tampering or loss, as well as the possibility of missing critical data due to the complexity of modern digital ecosystems.
2. ***Preservation:*** In the preservation phase, evidence is securely documented and stored to ensure its integrity and prevent tampering. A major challenge here is maintaining data integrity, especially in distributed systems, while avoiding contamination or alteration of evidence. Threats such as unauthorized access, data corruption, or encryption barriers imposed by advanced technologies can compromise the reliability of the preserved evidence, making secure handling protocols essential.
3. ***Analysis:*** The analysis phase focuses on applying forensic tools and techniques to extract, interpret, and analyze meaningful insights from the collected evidence. Challenges in this phase include dealing with large-scale data analysis, processing encrypted information, and navigating advanced anti-forensic techniques designed to obscure critical evidence. Additionally, the complexity of modern forensic tools can lead to overlooked evidence, posing a significant threat to the investigation's outcome.
4. ***Documentation:***: During the documentation phase, a detailed and transparent record of the investigation process, findings, and methodologies is compiled. Ensuring thoroughness in documentation is critical to meet legal admissibility requirements. Challenges arise when incomplete or unclear records fail to meet these standards, which can lead to the rejection of evidence in court. Furthermore, any mismanagement of this phase poses the threat of undermining the credibility and reproducibility of the forensic investigation.
5. ***Presentation:*** The presentation phase involves preparing evidence in a legally admissible format and presenting findings effectively for use in legal proceedings or organizational reviews. This phase requires translating technical findings into formats that are comprehensible to non-technical stakeholders, such as judges, juries, or organizational leaders. Challenges include presenting evidence clearly and persuasively while addressing potential threats of misinterpretation or insufficiently substantiated findings, which can weaken legal arguments or decision-making processes.

By systematically outlining these phases, the framework provides a cohesive understanding of digital forensics, bridging foundational concepts, subdomains, and investigative workflows. This structured approach not only enhances the integrity and reliability of evidence but also identifies critical points where issues and challenges may arise, ensuring robust and effective investigation processes. With this foundation, the next section focuses on previous studies, analyzing key tools, methods, and challenges, and identifying gaps that could guide future developments in digital forensics.

3. Related works

Certainly, by utilizing every aspect of cloud computing to achieve The field of digital forensics has evolved significantly, encompassing diverse subdomains that address distinct aspects of evidence identification, extraction, and analysis. Core areas, including network, mobile, database, IoT, cloud, and multimedia forensics, tackle unique challenges while sharing the common goal of ensuring data integrity and security. Over the years, these subdomains have seen significant advancements, with researchers proposing innovative methodologies, tools, and frameworks to address domain-specific issues. However, persistent gaps remain, requiring attention to strengthen the applicability and robustness of forensic techniques. Fig. 3 illustrates the digital forensics processing timeline, outlining the phases of Identification, Preservation, Analysis, Documentation, and Presentation, while highlighting the phases currently addressed and those that remain underexplored.

Building on the digital forensics processing timeline outlined in Fig. 3, Table 2 provides a structured analysis of how the six core domains align with these phases. This mapping not only illustrates the phases addressed and those that remain underexplored but also highlights the theoretical concepts driving advancements in each domain. To maintain clarity and avoid conceptual overload, Table 2 is presented separately from Table 1, as it focuses specifically on domain-level phase coverage rather than foundational study contributions. By offering this detailed perspective, Table 2 complements Fig. 3, enabling a clearer understanding of existing contributions and laying the groundwork for addressing critical research gaps.

Thus, the insights presented in Table 3 reinforce the need to explore each domain in greater detail to uncover the nuances of existing research, distinct contributions, and areas requiring further attention. The following subsections delve into the six core domains of digital forensics, examining how theoretical advancements and practical applications have shaped the field, while also identifying persistent challenges and opportunities for innovation.

3.1. Network forensics

Network forensics investigates network activities to uncover, analyze, and mitigate security breaches, ensuring the integrity of data transmissions and protecting against network-based crimes. Consequently in (Masinde and K Graffi, 2020), author introduces a peer-to-peer network model capable of integrating cyber forensics through the internet while a (Prasad and Rohokale, 2020; Y Qi, 2019) study mainly thrives on distributed techniques, offering an individualized approach to monitoring, evaluating, and retrieving network traffic data within a network.

An architecture is introduced for tracing and attributing network attacks by (Yang et al., 2022), that delves into common attacks, existing tools, and current developments in network techniques such as Honeypots, IP Traceback, and Intrusion Detection Systems. Packet analysis emerges as a fundamental traceback method, explored in (LF Sikos, 2020) which presents the comprehensive survey of packet analysis,

Table 2
Theoretical concepts and phase coverage across digital forensics domains.

Domain	Phases Covered	Missing Phases	Key Theoretical Concepts
Network Forensics	Identification, Preservation, Analysis, Documentation	Presentation	AI-driven methodologies, SDN-based processes, encryption challenges, vulnerability assessment tools.
Mobile Forensics	Identification, Preservation, Analysis, Documentation	Presentation	Secured device recovery, location-based techniques, advanced data extraction platforms.
Database Forensics	Identification, Preservation, Analysis, Documentation	Presentation	Blockchain trust models, AI for query analysis, proactive recovery approaches.
IoT Forensics	Identification, Preservation, Analysis, Documentation	Presentation	IoT-specific forensic models, AI for smart devices, blockchain integration.
Cloud Forensics	Identification, Preservation, Analysis, Documentation	Presentation	Multi-source evidence frameworks, blockchain for incident response, cloud forensic readiness.
Multimedia Forensics	Identification, Preservation, Analysis, Documentation, Presentation	–	AI-driven tampering detection, steganography techniques, facial biometric analysis.

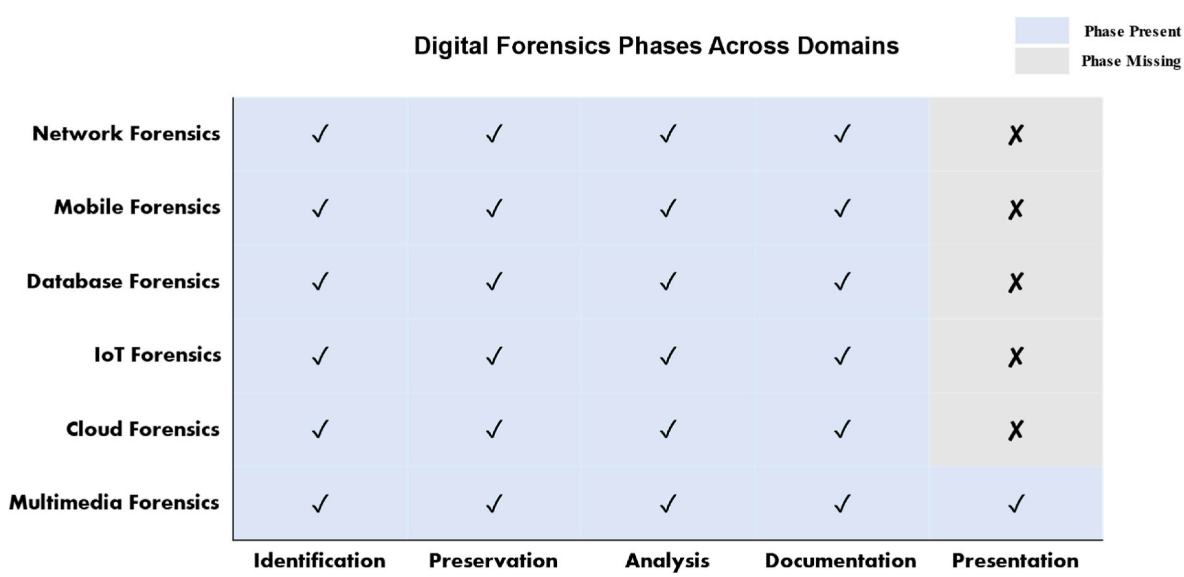

Fig. 3. Phases of digital forensics across six core domains.

Table 3
Mapping research overlaps, distinctions, and gaps across digital forensics core domains.

Domain	Contributions			Research Gaps Addressed
	Existing	Overlapping	Distinctive	
Network Forensics	Peer-to-peer network integration, distributed forensic models, multi-layer encryption methods (e.g., TOR, VPN).	Focus on encryption barriers, AI-powered traffic analysis, and tracing methodologies like Honeypots and IP traceback.	Extends focus on multi-encryption challenges, TOR/Proxies investigation, and distributed forensic tools for scalability.	Highlights limitations in real-time analysis, discusses forensic readiness in distributed environments, examines encryption-handling frameworks.
Mobile Forensics	Device recovery tools (e.g., Android/iOS), SIM card extraction, forensic hashing for secure evidence retrieval.	Emphasis on device-specific recovery techniques, data extraction tools, and cross-platform forensic methodologies.	Builds upon cross-platform synchronization and encrypted application challenges, proposing unified forensic models for complex architectures.	Identifies standardization issues in multi-device forensics, proposes unified models for encrypted apps and memory architectures.
Database Forensics	SQL/NoSQL forensic recovery tools, Oracle LogMiner for metadata and query analysis, blockchain readiness for data validation.	Investigations into SQL and Oracle databases with tools like LogMiner and focus on metadata recovery and query activity analysis.	Expands forensic approaches to NoSQL systems like MongoDB, integrating SQL and blockchain readiness strategies for decentralized systems.	Synthesizes SQL and NoSQL techniques, suggests strategies for forensic analysis in decentralized (blockchain-based) databases.
IoT Forensics	Cloud-based forensic models, fitness tracker forensic readiness, blockchain for audit trail scalability.	Conceptual models for IoT forensic readiness, AI-driven methods for evidence analysis, and blockchain audit trails.	Integrates blockchain-based audit trails with scalability challenges for heterogeneous IoT ecosystems, proposing holistic frameworks.	Reviews hybrid/multi-cloud readiness challenges, discusses blockchain-based traceability, analyzes cross-layer acquisition models.
Cloud Forensics	Smart contract-based evidence validation, forensic readiness models for cloud ecosystems, incident response frameworks.	Frameworks addressing forensic readiness for SaaS/IaaS models, blockchain for evidence immutability, and cloud log analysis.	Addresses multi-cloud and hybrid system readiness with proactive incident response and advanced blockchain-based management.	Proposes scalable forensic tools for emerging media (e.g., deepfakes, live streams), categorizes forgery detection gaps.
Multimedia Forensics	Forgery detection techniques (e.g., copy-move), deep-fake analysis, steganography-based authentication methods.	Digital watermarking, steganography analysis, and forgery detection techniques for images and videos.	Proposes frameworks for emerging formats like deepfakes, prioritizing scalability and standardizing multimedia forensic tools.	Detection of manipulations in emerging formats like deepfakes and real-time streams, with scalable cross-media tools.

including deep packet inspection, and reviews AI-powered methods for advanced network traffic classification and pattern identification (Waseem et al., 2021). Moreover, the resilience of network forensics is highlighted by its ability to conduct examinations even when the event's power is cut off in which memory itself stores crucial network data, encompassing public links, accessible ports, viewed websites, cookie files, and server records (Aslan et al., 2023).

Furthermore (Arshad et al., 2021), acknowledges the challenges posed by multi-encryption layer technologies, especially when dealing with TOR web browsers, proxies, and VPNs. In efforts to enhance the investigative processes of network forensics are evident in (IS Alansari, 2023) where the comprehensive model proposed in design science research comprises six defined processes and related activities, demonstrating flexibility and completeness when applied to various network types and legal frameworks. Similarly, a distributed network forensics framework presented in (Ghabban et al., 2021; Malvankar and A Jain, 2021; Moussa et al., 2019) underscores the significance of network forensic tools aiming to enhance the investigative process and strengthen the ability to respond quickly to digital crimes.

Moreover (MR Al-Mousa, 2022), contributes to the field by suggesting an approved data structure and theoretical model, offering theoretical concepts that can advance the construction and elucidation of actions related to incidents under investigation in the realm of forensics cybernation on an online circle network. Thus, network forensics emerge as critical subdomain of digital forensics especially in face of escalating network-based crimes along the integration of advanced techniques, frameworks, and architectures that ensure ongoing relevance in cybersecurity.

Building upon these advancements, this study combines insights into multi-encryption challenges and explores distributed forensic tools to enhance scalability and real-time analysis for evolving network crimes. By extending the discussion on encryption barriers and proposing strategies for distributed environments, this work addresses persistent gaps in forensic readiness and offers innovative approaches to network crime mitigation.

3.2. Mobile forensics

Mobile forensics focuses on retrieving, preserving, and analyzing evidence from portable devices, addressing the increasing use of mobile technology in modern investigations. Similarly, some cordless devices framed by BlackBerry (BB) were examined in (B Yankson, 2021) from the forensics perspective while an advanced tool named PDD was presented in (Kumar, 2020) for memorialization and forensics examination of gadgets that operate the PDAs Palm Operating System and various tools and techniques for PDAs, GSM, GPS, and portable mobile devices were suggested in (Ç Bakır, 2022).

A novel framework was presented in (Tayeb and C Varol, 2019) for the eradication of proof from main memory (such as ROM and flash memory) while some scientists recommended sim-data carver techniques to eradicate an entire filesystem for mobile devices, LINUX, and disk operating systems in (Buquerin et al., 2021). The retrieval of media, multimedia, and previously deleted evidence from mobile flash-derived memories was discussed in (Fukami and K Nishimura, 2019; Xie et al., 2022). Furthermore, the author (Humphries et al., 2021) introduced methodologies for cellphone control and mobile SIM programming. Meanwhile, in (Almehmadi and O Batarfi, 2019), the researcher proposed a mobile data acquisition method for Android and iOS devices, assessing its forensically sound nature, and mobile forensics hashing techniques were presented in (Chang et al., 2019).

A broad discussion of internal acquisition tools and logical extraction for mobile devices is proposed by an author in (Asim et al., 2019) while (Al-Faaruuq and DF Priambodo, 2022; Fukami et al., 2022; Vatsa et al., 2022) emphasizes the significance of Windows Mobile devices in forensic investigations by discussing the types of data recovery from smartphones and flash memory devices. Moreover (Kao et al., 2019), outlined a procedure for smartphone forensics under various scenarios, addressing the growth of the iOS operating system and (Keim et al., 2021) proposed a forensic analysis framework for Ubuntu Touch OS on PinePhone, utilizing autopsy. Physical data gathering techniques for secured passcode devices in Mobile Windows were discussed in (Boonkrong and T Heeptaisong, 2022). Meanwhile (Anwar et al., 2021), covered techniques for eradicating evidence from mobile GPS, focusing on Google's ability to track user location.

The author (Arshad et al., 2019; Sharma et al., 2020; Zhou et al., 2023) suggested an integrated time synchronization involving social network applications for forensic procurement and analysis with Android-powered devices. Logical and practical approaches for collecting information on the Sony Xperia XA2 were examined in (Tzvetanov and U Karabiyik, 2020). Moreover (Hort et al., 2021), introduced a technique and tools for physically obtaining data from unstable smartphone memories while (Salamh et al., 2021) provided recommendations for forensic examination of WhatsApp on Android phones. The study (H Alatawi et al., 2020; Sharanya et al., 2023) employed an illicit activity analyzing approach for drug trafficking and digital harassment, emphasizing smartphone forensic techniques.

From eradication methods to data extraction techniques, researchers have explored diverse aspects of mobile forensics. However, the survey identified a significant gap in mobile forensic research as existing work mostly focuses on technical procedures and problem-solving principles, neglecting the importance of a cohesive forensic paradigm and due diligence. The absence of unified models and standardized processes creates challenges for investigators, emphasizing the need for foundational guidelines in mobile forensics and urging future research to address these gaps.

To address these limitations, this study integrates device-specific strategies with emerging challenges such as encrypted applications and cross-platform synchronization. It also highlights the gaps in standardizing forensic methodologies for managing complex memory architectures and multi-device investigations. By proposing unified models, this work contributes to ensuring data integrity and improving the efficiency of mobile forensic techniques.

3.3. *Database forensics*

Database forensics specializes in uncovering and analyzing evidence within structured database systems, tackling challenges posed by multidimensional data environments and ensuring data integrity. Failures are often attributed to the complexity and diversity of DBMS, particularly when investigations focus solely on the filesystem. However, challenges arise when applying database forensics paradigms to database system investigations, as highlighted by authors in (Al-Dhaqm and Ikuesan, 2021; Al-Dhaqm et al., 2021a, 2021b; Alfadli et al., 2021; Chopade and VK Pachghare, 2019). Due to this, collaboration among digital investigators becomes crucial in analyzing databases effectively.

Most database forensic analyses in previous studies focus on recovering metadata and stored content, rather than on event reconstruction (Choi et al., 2021). Addressing this, researchers proposed an investigative process paradigm tailored for the Oracle Database, involving phases like stopping the database operation, gathering information, rebuilding the database, and repairing the database's integrity (Hassan et al., 2023). Furthermore (Alhussan et al., 2022), created the Oracle LogMiner program, which recreates operations when monitoring features are disabled. For this, multiple frameworks with a focus on Oracle databases have been proposed. The first approach explores using Oracle log files to uncover attacker activity (Kieseberg et al., 2019), while the second investigation provides a method for recovering deleted information in Oracle objects (Gedam and BB Meshram, 2021).

Another study (Mousa et al., 2020), utilizes audit trails and Listener files to capture threats targeting Oracle authorization processes, including IP addresses, user identifiers, and session details. Additionally (Flores Armas, 2019), emphasizes the need to examine various types of information, risks associated with data leakage, and technologies to ensure information security, encompassing physical protection, firewalls, encryption, and data monitoring. Addressing the absence of auditing, a detection investigation forensic model is proposed in (SKA Zakarneh, 2023), outlining how an examiner could identify evidence of information stealing in Oracle databases.

For SQL Server databases, researchers in (Wagner et al., 2019) propose a forensic analysis approach utilizing temporal tables and extended events where the four-phase DBFI model is employed for Database Forensic Investigation (DBFI). A framework for MySQL databases is introduced in (Nissan et al., 2023), labeled as the detecting discrepancies in the database schema, focusing on locating, labeling, and explaining bytes in the MySQL database system to provide a standard storage format (DB3F) and a related toolbox (DF-toolbox) for database forensic artifacts. Thus, supervised learning models, like support vector machines (SVM), are suggested in (Khanuja and Adane, 2020) to approximate recently executed queries based on forensic artifacts in memory, focusing on reverse-engineering database activity.

An improved forensic technique is proposed in (Aggarwal and Rehman, 2021), aiming to rebuild basic SQL Data Definition Language (DDL) commands while highlighting Data Manipulation Language (DML) statements, enhancing its utility by incorporating the crucial DDL statement that is useful in digital investigations, especially in unearthing illicit financial transactions through database audit records. Furthermore (Khan et al., 2023), explores MongoDB database forensics, providing a system for examining and recreating questionable activity in MongoDB databases emphasizing NoSQL databases' distinct qualities as they handle substantial amounts of unstructured data. Thus, the journey from uncovering fraudulent activities in databases to developing robust forensic models reflects the complexity of modern database systems and the innovative tools required for thorough investigation.

This study extends the scope of database forensics by incorporating forensic techniques for NoSQL databases, such as MongoDB, and integrating them with traditional SQL approaches. It also addresses the challenges posed by blockchain-based database systems, proposing strategies to enhance forensic readiness in decentralized environments. By bridging SQL and NoSQL forensic techniques, this work provides a comprehensive framework for addressing data integrity issues across diverse database architectures.

3.4. *IoT forensics*

IoT forensics focuses on analyzing evidence from interconnected devices, addressing challenges posed by distributed and heterogeneous ecosystems. For this, conceptual process models have been proposed in (Dawson and A Akinbi, 2021) to guide forensic investigations by emphasizing three main stages: Device-level forensics, Cloud-level forensics (involving devices like Bluetooth or ZigBee), and Network-level forensics.

An in-depth Digital Forensic Investigation (DFI) architecture was proposed in (Islam et al., 2019) to improve forensic effectiveness in IoT while reducing dependence on Cloud Service Providers (CSPs). The ubiquity of IoT devices especially in sensitive and civilian contexts is emphasized in (Li et al., 2019) for identifying, capturing, analyzing, and displaying forensic artifacts from IoT gadgets and associated infrastructure involving the Amazon Echo. Furthermore, the potential of IoT to enhance automation and real-time management of vital infrastructure is recognized in (Bakhshi, 2019). Specific devices, such as Fitbit Alta HR and Xiaomi Mi Band 2, are e examined in (Kang et al., 2020) for their forensic significance, highlighting the usefulness of gathered data in confirming victim actions or interrogator responses near an incident.

A comprehensive framework for Digital Forensic Readiness (DFR) is presented in (Kebande et al., 2020), leveraging the ISO/IEC 27043 international standard to enhance DFR capabilities and integrate forensic-by-design concepts into system architecture. A novel approach in (Scheidt and M Adda, 2020a) proposes identifying IoT devices by examining their device-specific "genes", forming a structure analogous to DNA. Additionally, new techniques for forensic investigative procedures and data exchange within a forensic setting were introduced in (Scheidt and M Adda, 2020b). The integration of blockchain in digital forensic records on IoT systems is explored in (Shakeel et al., 2021), where a Blockchain-Assisted Shared Audit Framework successfully identifies the origin and reason of data scavenging assaults within virtual resources (VR).

Another study focuses on creating an IoT testbed for forensic analysts, offering a forensic model to analyze relevant devices and potential sources of evidence (Hilgenberg et al., 2020). An in-depth investigation of the Google Home and Google Assistant apps on Android smartphones is proposed in (A Akinbi, 2020), extracting client-centric and cloud-native data remnants. The study uncovers key storage locations and deleted artifacts such as user accounts and voice data, offering valuable insights for forensic investigators. The overarching landscape of IoT forensics, with a specific emphasis on the use of artificial intelligence (AI), is explored in (Atlam et al., 2020). Moreover, (Gómez et al., 2021), presents a conceptual technique combining common elements of IoT devices into a comprehensive forensic framework for investigations.

Highlighting the importance of being prepared for digital forensics, especially in IoT forensics for businesses, a readiness framework is presented (Zulkipli and GB Wills, 2021) that identifies crucial aspects influencing IoT forensics research. Furthermore, a preventative cyber forensics method framework with honeypots for digital IoT probes is introduced in (Raman and Varadharajan, 2021). The dual impact of IoT on various industries and its associated security risks are acknowledged in (Lutta et al., 2021). The study examines how existing models apply to complex IoT environments, revealing theoretical and practical gaps that require more realistic forensic approaches.

Building on existing advancements, this study develops forensic methodologies for Infrastructure-as-a-Service (IaaS) and Software-as-a-Service (SaaS) models. It emphasizes the need for proactive incident response planning and advanced blockchain-based evidence management. By addressing gaps in forensic readiness across hybrid and multi-cloud systems, this work offers scalable and interoperable forensic frameworks tailored to modern cloud environments.

3.5. Cloud forensics

Cloud forensics addresses the complexities of collecting, analyzing, and preserving evidence from remote and distributed cloud environments, focusing on challenges like jurisdictional issues and scalability. Thus (Khanafseh et al., 2019; Purnaye and Kulkarni, 2022) highlight the distinctive challenges of cloud computing systems, including log unification, missing terms in service level agreements, and the intricacies of cloud investigations research trends and suggesting a taxonomy scheme on how this area is developing.

Moreover (Liu et al., 2019), present the LiveForen framework, which employs secure protocols, attribute-based encryption, and a fragile watermark for real-time data integrity verification while (Jayaraman and A Stanislaus Panneerselvam -, 2021) propose an architecture to ensure cloud-based healthcare data integrity, incorporating a novel public auditing protocol. Moving to forensic readiness of CSPs (Sanda et al., 2022), compares and describes forensic processes employed by leading CSPs like AWS, Microsoft Azure, and GCP to support investigations and establish standard operating procedures. In (AlSaed et al., 2022), the author proposes a method using Apache Spark to analyze large volumes of cloud data logs while (Kumari and AK Mohapatra, 2022) proposes a framework to improve investigation effectiveness by offering a trustworthy method for handling complex data collection.

Recognizing the dynamic and distributed nature of cloud environments (Mujahid, 2023), proposes a framework that emphasizes proactive incident response and legal awareness in cloud environments. Introducing the Cloud Evidence Tracing System (CETS) (Wu et al., 2022), offers a novel structure leveraging Cloud Service Provider (CSP) APIs to reduce resource usage, enhance efficiency, and support scalable forensic analysis for large-scale systems.

Furthermore, ((Pourvahab and G Ekbatanifard, 2019) proposes a digital forensic architecture for Infrastructure-as-a-Service cloud environments, utilizing Software-Defined Networking and Blockchain technology for improved evidence reliability, security, and integrity by employing Sensitivity Aware Deep Elliptic Curve Cryptography (SA-DECC), Fuzzy Smart Contracts (FSC), and a Secure Ring Verification Authentication (SRVA) scheme for decentralized and tamper-proof evidence collection while (Rana et al., 2023) suggests a decentralized approach using smart contracts on the Polygon blockchain to enhance the security, integrity, and accessibility of digital evidence.

To address evidence collection from virtual machine layers (Raju and G Geethakumari, 2019; Rane and Dixit, 2019), introduce a Cloud Forensic Readiness (CFR) model to enhance the availability of virtual machine snapshots for forensic investigations from the OpenStack Cloud Environment while (Awuson-David et al., 2021) introduces the Blockchain Cloud Forensic Logging (BCFL) framework to process transaction logs, maintain integrity through smart contracts, and ensure immutable logs across IaaS, PaaS, and SaaS models. Further enhancing Blockchain-as-a-service (BaaS) (W. Zheng et al., 2019), presents a framework to make cloud Blockchain adoption possible for organizations, preserving digital cloud assets with advanced capabilities in the cloud Platform-as-a-Service model.

Furthermore (Rane and Dixit, 2019), present the BlockSLaaS (Blockchain-assisted secure logging as-a-service) architecture for cloud environments to provide immutable log storage, offering fine-grained access control for Cloud-Forensics Investigators (CFI) and advancing cloud-based digital forensics procedures while (Nyaletey et al., 2019) presented the enhance data reliability and tracing, preserving data integrity and enabling efficient distributed data sharing via Blockchain-Interplanetary-Filesystem (BIPFS) method. Similarly (Arif et al., 2024), highlights the integration of blockchain technology with federated learning for decentralized, privacy-preserving threat detection and immutable logging in zero trust architectures, ensuring comprehensive security in distributed environments. Lastly (Lone and RN Mir, 2019), proposes Forensic Chain architecture, utilizing blockchain technology to maintain digital chain of custody and preserve evidence integrity throughout collection. Despite extensive research, a significant gap remains in cloud forensics literature, pointing to key future directions when it comes to conventional methods and technologies that are unique to cloud environment.

This study focuses on advancements in multimedia forensics by proposing a framework for cross-media forensic analysis that prioritizes scalability and robustness. Additionally, it addresses challenges such as detecting manipulations in emerging media formats like deepfakes and real-time content streams, offering pathways for standardizing tools and enhancing the reliability of multimedia forensic investigations.

3.6. Multimedia forensics

Multimedia forensics examines digital media to detect forgery, tampering, and authenticity issues, ensuring content integrity in forensic investigations. In a comprehensive overview of digital picture forgery detection techniques, the author (Bourouis et al., 2020; Costa et al., 2020; Saber et al., 2020) delves into various approaches, including digital watermarking, digital signatures, copy-move, image retouching, and splicing that help researchers understand the strengths and limitations of image forensics, comparing deep learning and CNNs with other detection techniques.

To tackle the pervasive problem of video and image tampering (Kaur and Jindal, 2020; L. Zheng et al., 2019), suggests a framework by providing an overview of tampering types, datasets, and detection techniques to differentiate between real and manipulated video and image data. Highlighting the importance of copy-move forgery detection (CMFD) methods, the author outlines a strategy to provide an accurate and efficient CMFD approach in (Ansari et al., 2020; Gupta et al., 2022; Teerakanok and T Uehara, 2019) emphasizing passive forgery, copy-move forensics, resampling, compression, and inconsistency — offering insights to scholars studying digital media manipulation and advancing research in digital forensics.

In (Chutani and Goyal, 2019), researchers suggest a framework focused on forensic steganalysis, including multi-class classification, payload size, key extraction, and performance indicators that address the need for deeper insights into steganographic information, contributing to a comprehensive understanding of steganography and steganalysis while the data-driven approach proposed in (Yang et al., 2020) summarizes data-driven algorithms in image source forensics, offering insights into visual data forensic applications. In the context of steganography and steganalysis (Dalal and Juneja, 2021), emphasizes real-world applications, cybercrime examples, and the impact on national security.

Addressing the significance of data science and digital forensic approaches in Child Sexual Abuse Material (CSAM) situations (Sanchez et al., 2019), proposes a methodology analyzing practitioners' views on the usefulness, efficiency, and precision of these tools in handling CSAM cases. In (Cifuentes et al., 2022), the author investigates and assesses various approaches to identifying sexually explicit material in movies, with an emphasis on deep learning-based methods, and highlights the need for precise pornography detection in the context of the rise in illicit content on the internet. In (Nowroozi et al., 2021), the author presents a methodology to improve the resilience of machine learning-based binary modification detectors in picture forensics while (Alsmirat et al., 2020; Shelke and Kasana, 2021) focuses on passive video forgery detection tools, discussing deepfake detection, anti-forensics, benchmark datasets, and a generalized method architecture.

To improve facial analysis and recognition systems (Becerra-Riera et al., 2019), presents a method that gives a thorough overview of facial identifiable visual attributes, emphasizing soft biometrics and how such traits are represented and categorized. A cutting-edge examination of nighttime visual refining techniques is provided in (M. Choi et al., 2022), to enhance low-light surveillance quality, critical for forensic and security monitoring. By developing methods to detect manipulation, tampering, or forgery in multimedia, this field contributes to enhancing data security and fostering confidence in the digital information ecosystem.

This study integrates advancements in multimedia forensics by proposing a framework for cross-media forensic analysis that prioritizes scalability and robustness. Additionally, it addresses challenges such as detecting manipulations in emerging media formats like deepfakes and real-time content streams, offering pathways for standardizing tools and enhancing the reliability of multimedia forensic investigations.

Additionally, to consolidate the extensive literature reviewed in Sections 3.1 to 3.6, Table 3 presents a synthesized mapping of contributions across six core digital forensics domains. Each listed contribution has been systematically derived from individually analyzed studies, as detailed in the corresponding domain discussions. Rather than listing each paper again, Table 3 aggregates recurring patterns and distinctive insights to offer a comparative view of existing, overlapping, and unique advancements. This ensures that the table remains concise while still grounded in the comprehensive, paper-specific analysis already presented. The final column further highlights the research gaps directly addressed in this paper, offering a clearer view of how this work advances the field.

Thus, this exploration of digital forensics across six core domains provides a structured understanding of the advancements achieved and the challenges that persist. Notably, gaps such as the missing presentation phase in five domains and the limited application of theoretical frameworks, including blockchain-based forensic readiness and IoT-specific methodologies, underscore critical areas for further research. Addressing these challenges will not only refine existing forensic techniques but also foster innovative approaches to enhance the adaptability and effectiveness of digital forensic practices, ensuring their continued relevance in an increasingly complex technological landscape.

4. Recent security issues, techniques and tools in digital forensics

The dynamic field of digital forensics requires ongoing refinement and development of security tools and procedures due to the ever-evolving cyber threats. Diverse subdomains within digital forensics have evolved because firms struggled with the sophistication of cyber-attacks, each requiring specific tactics for best security as demonstrated in Fig. 4. The toolkit is vast and includes network forensics, mobile forensics, database forensics, IoT forensics, cloud forensics, and multimedia forensics. Such tools are made to extract and examine digital evidence, providing detectives with important information about cyberattacks. Keeping up with technical developments is essential to protecting sensitive data and preserving the integrity of digital environments in this ever-increasing conflict between cyber-criminals and defenders.

Fig. 4 outlines the diverse subdomains within digital forensics, showcasing the breadth of issues, techniques, and tools employed to address the challenges posed by evolving cyber threats. While some techniques address overarching concerns that span multiple domains, others are uniquely tailored to specific forensic environments, such as IoT, mobile, or cloud forensics. To provide a clearer understanding of these distinctions, Table 4 categorizes the key issues, techniques, and tools into generic approaches applicable across all domains and domain-specific solutions tailored to particular contexts. Furthermore, the table incorporates the five phases of the digital forensics processing timeline: Identification, Preservation, Analysis, Documentation, and Presentation, mapping these techniques and tools to the relevant stages of the investigation workflow. This combined classification highlights the relevance, scope, and role of each method, offering a comprehensive perspective on their applications in digital forensics (see Tables 5–10).

The classification presented in Table 4 not only highlights the versatility of digital forensic tools and techniques but also underscores the importance of addressing domain-specific challenges alongside generic frameworks. By mapping these methods to the phases of the digital forensics' timeline, the table provides a dual-layered roadmap for practitioners, ensuring a more structured and actionable understanding of their applications. While foundational techniques like network log analysis are broadly applicable, specialized methods such as IoT device imaging or mobile app reverse engineering remain essential for tackling unique challenges within specific contexts. As digital forensics continually evolves to counteract the sophistication of modern cyber threats, leveraging tailored investigative approaches ensures the reliability, integrity, and applicability of forensic processes. The following sections delve deeper into these subdomains, exploring their unique issues, advancements, and critical roles in modern cybersecurity.

4.1. Network

4.1.1. Traffic capture and analysis

By offering insights into network activity and spotting possible security concerns, traffic collection and analysis play a critical role in cybersecurity. However, this process faces several challenges, including encrypted network traffic, which obscures data content, and the overwhelming volume of network activity, which complicates real-time threat detection. Additionally, sophisticated attack methods like traffic spoofing and protocol tunneling can evade conventional analysis methods, requiring more advanced detection strategies. Packet sniffers are a popular tool for recording network traffic because they can intercept and collect data packets as they move across the network (D Saputra, 2019; LF Sikos, 2020). Another useful tool is a network probe (Mishra et al., 2021), which actively explores the network to get information about its security and performance. These issues necessitate the use of innovative tools and methods to ensure effective monitoring, while enabling cybersecurity experts to understand how people communicate within a network using these techniques.

To find an unusual activity that can point to security breaches, recorded traffic analysis is important. Network traffic patterns may be carefully examined to identify malware infestations, data breaches, and unwanted access attempts. For traffic analysis, tools like Splunk,

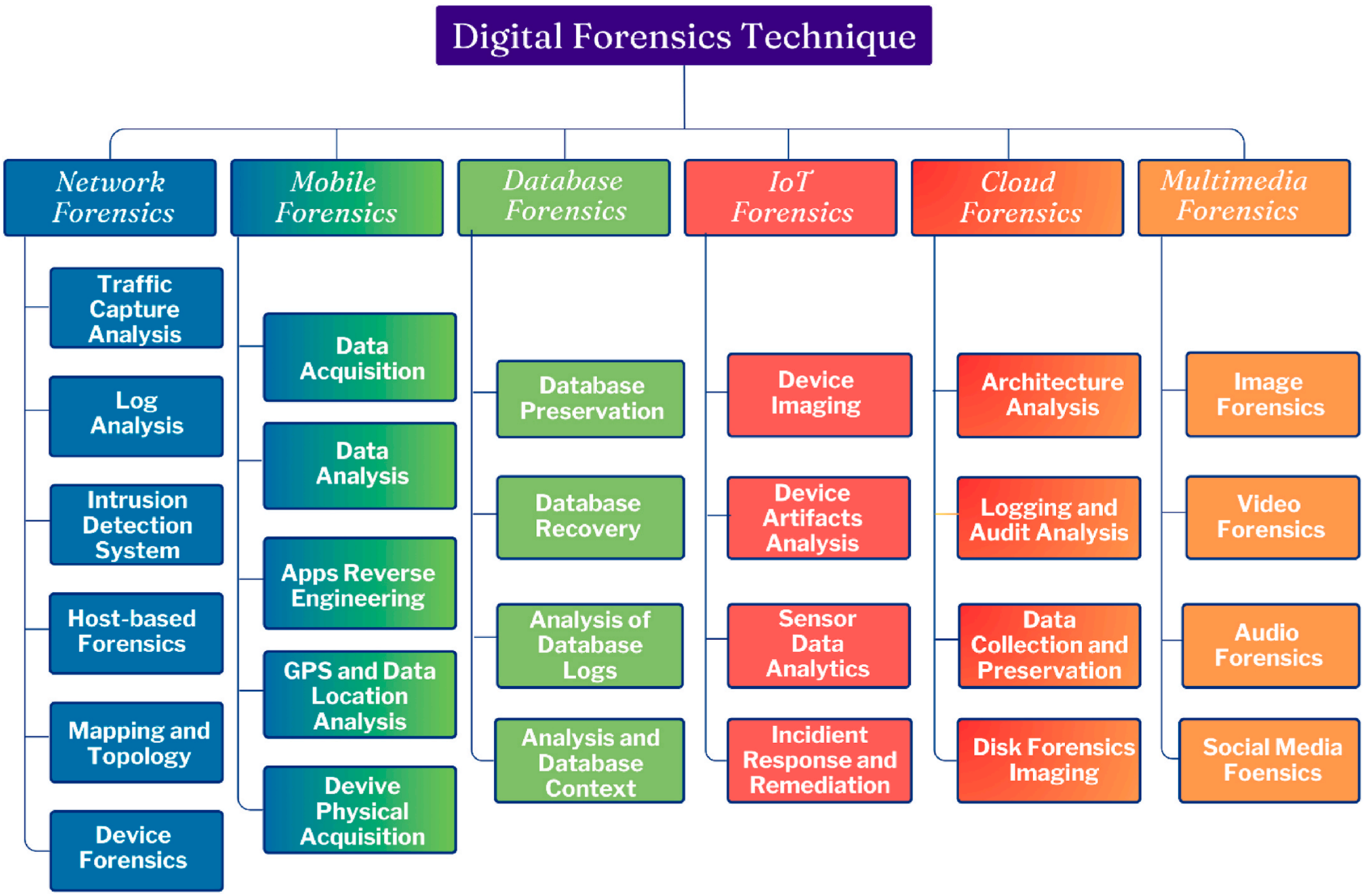

Fig. 4. Decoding the intricacies of cyber attacks by exploring security solutions in digital forensics landscape.

tcpdump, and Wireshark tend to be employed (Musa et al., 2019; Sha et al., 2021). For example, Wireshark enables thorough packet content analysis, while tcpdump is a command-line packet analyzer. In contrast, Splunk is a feature-rich platform that facilitates real-time monitoring and log analysis, which helps identify abnormalities and possible security problems (Stelly and V Roussev, 2019). With the help of these tools and methods, cybersecurity experts can proactively protect networks from a wide range of threats.

4.1.2. Network log analysis

A key component of forensic investigations is network log analysis, which offers an enormous amount of data to uncover the specifics of security events and possible breaches. However, the process is not without its challenges. Logs from diverse sources, such as firewalls, routers, and DNS, are often in varying formats, making aggregation and correlation complex. The immense volume of logs generated in modern networks adds to the difficulty, particularly when distinguishing between benign anomalies and actual security threats. Furthermore, advanced persistent threats (APTs) may leave subtle traces that require sophisticated detection techniques. By examining network logs, cybersecurity experts may determine the series of actions that led to a security incident, which aids in determining the extent, significance, and source of the issue that occurred. Advanced AI and machine learning techniques have further enhanced the ability to detect anomalies and patterns within large-scale log data, providing deeper insights into potential threats (Dunsin et al., 2024; R El-Kady, 2025). These innovations are crucial to overcoming the challenges outlined, enabling organizations to improve their security posture and streamline incident response processes. This forensic method helps with handling incidents and makes it easier to create plans that effectively stop security risks in the future.

Diverse kinds of network logs are essential sources for detectives. Firewall logs provide information about traffic that the firewall permits or blocks and may be used to identify attempted unauthorized access. Router logs include information about connection problems, routing modifications, and network traffic. Communication with malicious sites can be uncovered using DNS logs, which record requests for domain name resolution. Log analysis requires specific tools and methods to be done effectively. Log data aggregation, processing, and visualization are frequent uses for platforms such as Elastic Stack, Graylog, and Squert (Kotsiuba et al., 2019). With the use of these technologies, cybersecurity experts can quickly and effectively go through enormous volumes of log data, see trends, and spot abnormalities that can point to security events. With these tools, organizations may enhance their capacity to rapidly address security risks and strengthen their network defenses.

4.1.3. Network intrusion detection system

Through active monitoring and analysis of network traffic, Network Intrusion Detection Systems (NIDS) play a critical role in strengthening cybersecurity by identifying and alerting users to hostile actions. Despite their effectiveness, NIDS face significant challenges. These include difficulty detecting encrypted malicious traffic, the risk of high false positive rates in anomaly-based systems, and resource-intensive requirements for real-time packet inspection in large-scale networks. Additionally, sophisticated attack techniques, such as polymorphic malware, can evade signature-based detection by continually altering their code. Real-time packet inspection and comparison against a predetermined set of rules or signatures that indicate suspected risks are how these systems function. Signature-based NIDS offers strong protection against known threats by searching for certain patterns or signatures linked to established attack techniques. Anomaly-based network

Table 4
Differentiating generic and domain-specific issues, techniques and tools in digital forensics.

Domain	Techniques/Tools	Generic or Domain-Specific	Phase in Timeline
Network	Traffic Capture, Log Analysis, Intrusion Detection, Host-based Forensics, Mapping and Topology, Device Forensics	Generic (All Domains)	Identification, Analysis
Mobile	Data Acquisition, Data Analysis, Apps Reverse Engineering, GPS and Data Location Analysis, Device Physical Acquisition	Domain-Specific (Mobile)	Identification, Preservation, Analysis, Documentation
Database	Preservation, Recovery, Logs Analysis, Content Analysis	Domain-Specific (Database)	Preservation, Analysis, Documentation
IoT	Device Imaging, Artifacts Analysis, Sensor Data Analytics, Incident Response and Remediation	Domain-Specific (IoT)	Identification, Preservation, Analysis, Documentation
Cloud	Architecture Analysis, Logging and Audit Analysis, Data Collection, Disk Forensics Imaging	Domain-Specific (Cloud)	Identification, Preservation, Analysis
Multimedia	Image Forensics, Video Forensics, Audio Forensics, Social Media Forensics	Domain-Specific (Multimedia)	Identification, Analysis, Documentation

intrusion detection systems concentrate on identifying departures from predetermined standards of typical network activity. Activities that drastically depart from the usual patterns are flagged by these systems, which may indicate the presence of novel and unidentified risks. Hybrid NIDS provides a thorough defensive scheme by combining the strengths of both signature and anomaly detection techniques (Khraisat et al., 2019). To address these challenges, integrating artificial intelligence (AI) and machine learning has proven beneficial, enabling systems to better adapt to evolving threats and reduce false positives while maintaining robust defense capabilities.

Network intrusion detection systems (NIDS) are essential tools for network forensics investigations because they help locate and address security events. NIDS possesses the capacity to identify potentially harmful patterns or behaviors in network traffic that might indicate an intrusion or security violation. The integration of artificial intelligence (AI) into cybersecurity systems has significantly enhanced threat detection and response capabilities, as highlighted in (Tyagi et al., 2025). While AI models are not limited to specific tools, their application in analyzing large-scale network traffic data has contributed to more efficient anomaly detection and adaptive defense mechanisms (Shirbhate and SR Gupta -, 2024). When suspicious or malicious behavior is found, the NIDS produces logs or alerts that offer important details for additional investigation and action. By taking a proactive stance, cybersecurity experts may quickly identify and eliminate any threats, enhancing the network's overall resilience. In addition to being a preventive measure, NIDS is a useful tool for post-incident analysis, helping to reconstruct events and create plans to improve privacy in the decades to come.

4.1.4. *Network host-based forensics*

To comprehend more regarding security events, network host-based forensics systematically gathers and examines forensic data from specific network hosts (Qureshi and Tunio, 2021). However, several challenges arise in this process, such as ensuring the integrity of volatile data during collection, overcoming encryption barriers on stored data, and handling large-scale environments with numerous interconnected hosts. Additionally, the risk of inadvertently altering evidence during analysis necessitates strict adherence to forensic procedures. Identifying and isolating the afflicted host or hosts is usually the first step in the approach. After the host has been isolated, forensic experts use specific tools and techniques to gather and examine data from its memory, storage, and other relevant sources. To shed light on the type and extent of the assault, this investigation attempts to reconstruct the series of events that culminated in a security issue.

In host-based forensic analysis, a range of technologies are employed to efficiently scrutinize and retrieve evidence. For instance, the potent open-source memory investigations tool Volatility makes it possible to retrieve data from a system's volatile memory. Mandiant's Redline is an additional tool for host-based investigations that includes file and memory analysis capabilities. One popular digital forensic tool is EnCase, which offers extensive features for gathering and examining evidence from a variety of sources, such as hard disks, memory, and network captures (Dweikat et al., 2020). These techniques aid in the

Table 5
Comparative analysis of techniques and tools within network forensics.

Subdomain	Technique/Tool	Key Features	Advantages	Limitations
Traffic Capture and Analysis	Wireshark, Tcpdump, Splunk, Packet sniffers, Network probes	Packet-level analysis, command-line packet analyzer, real-time traffic/log analysis, intercept/collect data packets	In-depth packet content analysis, lightweight/efficient, scalable anomaly detection, comprehensive traffic monitoring, identifies vulnerabilities	Struggles with encrypted traffic, require expertise, high resource usage, ineffective against encrypted traffic
Network Log Analysis	Elastic Stack, Graylog, Sguil, DNS logs, Firewall logs	Log aggregation and visualization, centralized log management, log visualization, domain name monitoring, traffic monitoring	Open source/scalable, user-friendly, real-time insights, detects malicious domains, identifies unauthorized access	Complex setup for Elastic Stack, lacks robust analytics capabilities, DNS tampering issues, overwhelming data from manual traffic logs
Network Intrusion Detection Systems (NIDS)	Signature-based NIDS, Anomaly-based NIDS, Hybrid NIDS, AI/ML-Enhanced NIDS	Matches known signatures, detects deviations from normal behavior, combines signature/anomaly detection, adaptive detection	Effective for known threats, detects evolving threats, comprehensive coverage, reduces false positives/scales for large datasets	High false-positive rates, resource-intensive/complex implementation, dependent on training data quality
Network Host-Based Forensics	Volatility, Mandiant's Redline, EnCase	File/memory analysis, comprehensive host forensic analysis, network discovery/port scanning	Supports host-based investigations, examines wide variety of digital evidence, easily debugs network infrastructure	Less effective in large-scale environments, expensive, requires expertise
Network Mapping and Topology	Nmap, Zenmap, SolarWinds Network Topology Mapper	GUI for Nmap, automatic network mapping	Simplifies visualization and network mapping, identifies hidden devices/connections	Limited analysis beyond topology, costly/resource-intensive
Network Device Forensics	Cisco IOS Forensic Toolkit, Juniper Secure Analytics	Forensic analysis of Cisco routers, log correlation for Juniper network devices	Extracts Cisco configurations, real-time anomaly detection	Device-specific, limited compatibility

Table 6
Comparative analysis of techniques and tools within mobile forensics.

Subdomain	Technique/Tool	Key Features	Advantages	Limitations
Mobile Data Acquisition	Oxygen Forensic Suite, Cellebrite UFED, GrayKey, Cloud backups, Physical extraction	Data processing, logical/physical extraction, passcode bypass, accessing remote data storage, hardware-level acquisition	Comprehensive extraction, encryption bypass, efficient for encrypted devices, retains data integrity, captures allocated and unallocated space	High resource usage, limited device support, specialized expertise required, legal/privacy compliance complexities, risk of data damage
Mobile Data Analysis	XRY, MOBILedit, Autopsy	Advanced extraction/analysis, timeline display, keyword searching, open-source timeline analysis	Extensive support for mobile devices, effective parsing, accessible and flexible for investigators	High licensing costs, limited non-mainstream support, training required
Mobile Apps Reverse Engineering	App usage trends analysis, IDA Pro, Apktool, JADX	Extracts/interprets user activity, disassembly/debugging, decompiles/reassembles APKs, converts bytecode into readable form	Reveals behavioral insights, in-depth functionality insights, effective resource/manifest analysis, simplifies app reverse engineering	Overlooks proprietary data formats, expensive, requires expertise, limited to Android apps, focus on static analysis
GPS and Data Location Analysis	Dynamic analysis, Google Maps Timeline, Geospatial formats (e.g., KML)	Executes apps in controlled environment, GPS visualization, standardized location representation	Monitors behavior in real-time, easy movement visualization, simplifies mapping and analysis	Ineffective for apps with anti-debugging, GPS signal inaccuracies, complex formatting requirements
Mobile Device Physical Acquisition	Visualization tools, JTAG imaging, Chip-off methods, FTK Imager	Trend/anomaly detection, direct memory access, hardware chip extraction, comprehensive data imaging	Identifies travel patterns, retrieves low-level data, accesses locked devices, supports forensic integrity	Differentiating benign from malicious activities is challenging, physical access and specialized hardware required, device damage risks, time-consuming

Table 7
Comparative analysis of techniques and tools within database forensics.

Subdomain	Technique/Tool	Key Features	Advantages	Limitations
Database Preservation	EnCase Forensic Imager, FTK Imager, Logical backups	Reliable forensic duplication, physical/logical imaging, SQL dumps	Ensures data immutability, maintains evidentiary integrity, facilitates selective duplication	High resource demand for large databases, limited scalability for massive datasets, risk of excluding critical unstructured data
Database Recovery	Recuva, PhotoRec, Advanced Data Recovery	Restores deleted files, file carving, recovery tools	Effective for standard file systems, handles fragmented/deleted data, recovers corrupted storage	Limited capabilities for encrypted data, inefficient for large datasets, high learning curve for advanced functionalities
Database Log Analysis	Splunk, Elasticsearch, Kibana	Log aggregation, scalable management, visualization/anomaly detection	Real-time insights, efficient for distributed log analysis, simplifies pattern recognition	Requires significant computational resources, configuration complexities, limited without robust data aggregation tools
Database Content Analysis	SQL editors, Statistical software, Data visualization tools	Structured queries, identifies trends, graphical representation	Enables precision in content analysis, facilitates detailed interpretation, simplifies complex structures	Requires deep knowledge of schemas, may overlook relationships in unstructured data, ineffective for fragmented datasets

Table 8
Comparative analysis of techniques and tools within IoT forensics.

Subdomain	Technique/Tool	Key Features	Advantages	Limitations
IoT Device Imaging	FTK Imager, EnCase Forensic Imager, JTAG Imaging	Comprehensive imaging, tailored imaging, memory extraction	Ensures data integrity, supports volatile/non-volatile memory, provides low-level access	Challenging for proprietary architectures, time-intensive for large datasets, risk of physical damage during disassembly
IoT Device Artifacts Analysis	IDA Pro, Binary Ninja, Firmware Analysis Tools	Firmware disassembly, dynamic/static analysis, reverse engineering	Identifies malicious code, highlights vulnerabilities, pinpoints backdoors	High expertise required, limited scalability for large networks, difficulty interpreting obfuscated firmware
IoT Sensor Data Analytics	Python Packages, Geospatial Tools, Statistical Software	Data manipulation, location-based analysis, trend identification	Streamlines pattern detection, enables context-specific insights, facilitates comprehensive analysis	Heterogeneous data complicates workflows, limited resolution for low-quality sensor data, may overlook real-time threats
IoT Incident Response and Remediation	Vulnerability Scanners, Network Monitoring Tools, IoT Security Platforms	Identifies misconfigurations, real-time surveillance, centralized detection	Reduces attack surface, detects potential threats proactively, strengthens overall IoT security	Requires frequent updates, resource-intensive for large environments, scalability issues in heterogeneous environments

Table 9
Comparative analysis of techniques and tools within cloud forensics.

Subdomain	Technique/Tool	Key Features	Advantages	Limitations
Cloud Architecture Analysis	Cloud Provider Documentation, Security Evaluation Tools, Visualization Tools	Architecture details, risk assessment, cloud mapping	Facilitates understanding of configurations, identifies weaknesses, simplifies architecture analysis	Limited standardization, may not cover all cloud layers, lacks scalability for dynamic environments
Cloud Logging and Audit Analysis	Splunk, LogRhythm, Cloud-Specific Log Analysis Tools	Log aggregation, tailored cloud log analysis	Enables real-time monitoring, strengthens security posture, ensures compliance	Resource-intensive for large environments, may require platform integration, limited interoperability
Cloud Data Collection and Preservation	Specialized Forensic Platforms, Legal Hold Features, Data Transfer Technologies	Evidence extraction, data integrity, secure transfer	Ensures compliance, preserves evidence, ensures chain of custody	Challenges with encrypted/deleted data, complex regulations, time-consuming for large datasets
Disk Forensics Imaging	Cloud Imaging Tools, Memory Acquisition Methods, Forensic Applications	Disk imaging, memory extraction, storage analysis	Preserves allocated/unallocated space, provides snapshots, facilitates detailed analysis	Difficult for ephemeral instances, requires CSP cooperation, scalability issues for complex systems

Table 10
Comparative analysis of techniques and tools within multimedia forensics.

Subdomain	Technique/Tool	Key Features	Advantages	Limitations
Multimedia Image Forensics	EXIF Analysis, Copy-Move Forgery Detection, Error Level Analysis	Metadata examination, duplication detection, compression anomaly detection	Identifies manipulation history, detects localized manipulations, reveals potential edits	Limited with missing/altered metadata, ineffective for large-scale alterations, prone to false positives in degraded images
Multimedia Video Forensics	Metadata Analysis, Compression Artifact Detection, Scene Consistency Analysis	Video creation history, tampering identification, event coherence analysis	Verifies video authenticity, detects frame-level edits, discovers tampered sequences	Limited with incomplete metadata, requires high computational resources, ineffective for low-resolution videos
Multimedia Audio Forensics	Spectral Analysis, Speaker Identification, Noise Reduction Techniques	Frequency visualization, voice pattern matching, clarity enhancement	Detects anomalies/splicing, matches voices to sources, improves analysis accuracy	Limited in noisy recordings, challenging in encrypted/mixed audio, may alter original evidence
Social Media Forensics	Sentiment Analysis, Social Media Archiving, Link Analysis Tools	Emotional tone interpretation, post collection, connection identification	Provides contextual understanding, preserves evidence integrity, highlights network dynamics	Limited for detecting manipulations, challenging with real-time deletions, difficult with vast data volumes

recognition and gathering of proof concerning malware infestations, data breaches, and further network-based assaults. Cybersecurity experts may adopt improved security measures and respond more effectively by using host-based forensics to obtain a thorough understanding of the strategies, methods, and tactics used by attackers.

4.1.5. Network mapping and topology

Gaining an understanding of network topology is essential for forensic investigations since it offers a fundamental understanding of how devices are connected inside a network. However, network mapping faces challenges such as incomplete or outdated topology information, dynamically changing network environments, and hidden devices that might not respond to discovery tools. These issues complicate accurate assessment of potential compromise sites and the understanding of attack vectors. Understanding network architecture helps investigators understand data flow, detect suspected compromise sites, and assess the effect of security breaches. It serves as the cornerstone for efficient incident response, enabling cybersecurity experts to deliberately allocate resources and countermeasures to control and reduce vulnerabilities.

To view and record network topology, a variety of tools and methods are used in network mapping. For active network discovery, Nmap and its graphical user interface Zenmap are frequently utilized, offering insights into live hosts, services, and their connections (Pansari and A Agarwal, 2020). An additional tool for automatically mapping networks that makes it possible to see the links between devices and the network architecture is SolarWinds Network Topology Mapper (Roy et al., 2022). These technologies enable cybersecurity experts to draw intricate network maps that are useful resources for locating possible attack points. Network maps assist investigators in identifying potential avenues of entry, comprehending potential assault routes, and formulating well-informed security plans for vital resources. Network mapping essentially plays a crucial role in the forensic procedure by strengthening a network's resilience and proactive protection against online attacks.

4.1.6. Network device forensics

Forensic evidence from important network components, such as routers, switches, and firewalls, is systematically collected and examined as part of network device forensic analysis. This process is often complicated by challenges such as ensuring the integrity of log data during extraction, dealing with vendor-specific configurations that vary widely between devices, and managing large volumes of historical data. Additionally, encryption and access restrictions on some devices can delay evidence collection and increase complexity, obstructing timely investigations. Finding the precise device or devices that are pertinent to the inquiry is the first step in the procedure. After that, forensic experts gather data from these devices, concentrating on records such as device logs, configuration files, and history data. Through the analysis of this data, investigators can get insights into the activities and modifications occurring within the network and find proof of attempted illegal access, modifications to device configurations, and possible network assaults. However, network device forensic analysis faces significant challenges, such as ensuring data integrity, handling diverse device configurations, and effectively analyzing log and configuration files. These limitations highlight the need for advancements in forensic tools and methodologies to enhance reliability and scalability in forensic investigations (Khan et al., 2022).

To extract and understand pertinent data, device forensic analysis uses specialized tools and methodologies. The Cisco IOS Forensic Toolkit is intended to analyze Cisco routers through the extraction of log files, configuration files, and memory contents. With capabilities for log analysis and event correlation, Juniper Secure Analytics is an additional tool that helps with the forensic examination of Juniper devices (Kaur and Jindal, 2020). These technologies aid in locating and gathering data that is essential to comprehending network device security posture. Investigators can identify abnormalities, determine the source of security problems, and determine whether illegal changes or attempts to gain access have taken place by carefully examining device logs and configurations. By addressing these challenges and improving forensic tools, organizations can strengthen their resilience against emerging cyber threats and ensure the reliability of their forensic investigations. While the tools and techniques discussed in the earlier sections address various aspects of network forensics, a comparative overview can further elucidate their distinct capabilities and limitations. To provide a clearer understanding of how these tools complement each other and tackle specific challenges in different subdomains, the table offers a detailed comparison.

This comparative table highlights the diverse range of tools and techniques available for network forensics. Each tool has its strengths and limitations, making it crucial to select the appropriate methodology based on specific investigative requirements and the nature of the challenges encountered. By understanding these distinctions, cybersecurity professionals can make more informed decisions, optimize their forensic processes, and enhance overall network resilience.

4.2. Mobile

4.2.1. Mobile data acquisition

Mobile data acquisition is the process of obtaining data from mobile devices using a variety of techniques, each with specific purposes. However, this process faces several challenges, such as bypassing security mechanisms like encryption and passcodes, ensuring the integrity of temporary data, and navigating the diversity of operating systems and device configurations. Cloud services add further complexity by requiring compliance with legal and privacy regulations when accessing remotely stored data. Physical extraction ensures a thorough capture of data by taking information straight from the hardware of a device. A less invasive method is offered by logical extraction, which concentrates on obtaining user data and system files via software interfaces. Accessing and obtaining data from cloud services linked to a mobile device is a component of cloud backups. It is crucial to preserve a secure chain of custody during the whole procurement process to guarantee the authenticity and admissibility of the gathered evidence in court.

Many specialized technologies and methods are used to make mobile data collecting easier. Notable examples are Oxygen Forensic Suite, Cellebrite UFED, and GrayShift GrayKey (Morin Da Silveira et al., 2020). Cellebrite UFED is well known for its ability to handle a large variety of devices and execute both logical and physical extractions. With its expertise in data processing and analysis, Oxygen Forensic Suite provides insights into user behavior and device interactions. GrayShift GrayKey is very good at getting around device passcodes so that data extraction is done quickly. These instruments are essential to forensic investigations because they help analysts retrieve critical data from mobile devices while following rigorous guidelines and preserving the chain of custody.

4.2.2. Mobile data analysis

A crucial step in digital forensics is mobile data analysis, which involves looking through retrieved data to find proof of illegal conduct. However, this process is not without its challenges. Investigators must deal with proprietary data formats, encryption barriers, and the rapid evolution of app ecosystems, which can render traditional tools ineffective. Furthermore, the large volume of data retrieved from mobile devices can complicate identifying relevant evidence in a timely manner. The risk of overlooking hidden or deleted data adds an additional layer of complexity. Data of all kinds, including text messages, phone records, images, videos, and app usage trends, are analyzed by investigators. Text messages and phone logs can provide important insights about communication patterns, connections, and timeframes; multimedia files, such as images, and videos, can act as visual proof. Since mobile applications frequently store data in proprietary forms that call for specific tools for efficient analysis, it is imperative to understand the subtleties of these data structures and formats. This information is essential for interpreting data from apps, revealing user activities, and finding any encrypted or concealed data that may be important for an investigation.

In order to handle the complexity of various devices and apps, a variety of tools and approaches are used in mobile data analysis. XRY, MOBILedit, and Autopsy are well-known tools in this field (Hamad and Eleyan, n.d.; V Fernando, 2021). With capabilities like timeline analysis and keyword searches, Autopsy is an open-source digital forensics software that facilitates the examination of data from mobile devices. One commercialized forensics tool that is well-known for providing extensive support for mobile device extraction and analysis is referred to as XRY. Another program that focuses on the analysis of data from mobile devices is identified as MOBILedit, and it has features like timeline display and data parsing. With the help of these technologies, investigators can effectively sort through enormous volumes of mobile data, identifying and interpreting information that is essential for digital forensic investigations.

4.2.3. Mobile apps reverse engineering

A key component of cybersecurity and digital forensics is mobile app reverse engineering, which is the act of breaking down and comprehending the internal operations of mobile applications. However, this process faces several challenges, including the presence of obfuscation techniques designed to conceal app logic, legal and ethical concerns related to reverse engineering, and the immense complexity of modern apps with extensive dependencies and third-party integrations. Furthermore, dynamic analysis may fail when apps implement anti-debugging measures, while static analysis can be hindered by encrypted or compressed code segments. This technique is used to reveal an app's functionality, spot any security flaws, and unearth potentially harmful or hidden features. App reverse engineering is done using a variety of techniques, including disassembly, dynamic analysis, and static analysis. While dynamic analysis runs the application in a controlled environment to monitor its behavior, static analysis looks at the code of the application without running it. The process of transforming an application's machine code back into a readable format for humans is known as disassembly, and it offers insights into the logic and organization of the program.

The field of mobile app reverse engineering uses a variety of tools and methods. The popular disassembler and debugger IDA Pro helps with mobile app assembly code analysis. Apktool makes it easier to examine resources and manifest files by enabling the deconstruction and reassembly of Android app packages (APKs). In contrast, JADX is an Android app that converts bytecode into a format that is easier for people to understand (Jo et al., 2019). These technologies are essential for breaking down mobile apps, giving forensic analysts and security experts the ability to evaluate the behavior of the app, spot security flaws, and enhance mobile app security.

4.2.4. GPS and data location analysis

In digital forensics and investigations, analyzing GPS and location data taken from mobile devices is essential since it may reveal important information about a person's movements and point out important locations. However, several challenges complicate this process. GPS data may be incomplete or inaccurate due to signal interference, device settings, or spoofing attempts by malicious actors. Additionally, ensuring the authenticity and reliability of location data is critical, particularly in legal contexts where evidence must meet strict admissibility standards. This kind of research builds a chronological record of an individual's whereabouts by analyzing timestamped location data. Based on common places, investigators can create linkages between people, follow journeys, and identify patterns of visiting. This information is especially useful in situations when it's critical to comprehend travel patterns, such as criminal investigations and cases involving missing people.

GPS and location data interpretation requires an understanding of geographic data formats and the use of visualization tools. Geographical characteristics and their qualities can be represented in a standardized manner using geospatial data formats like Keyhole Markup Language (KML). Investigators can map and analyze location data using visualization tools, such as Google Maps Timeline, to gain a better picture of the subject's travels over time. Investigators can find trends, patterns, and anomalies that aid in the whole investigation process by superimposing GPS data onto maps and using visualization tools (Mirza and U Karabiyik, 2021). By offering a thorough perspective of location data, these technologies improve the investigation process and make it easier for investigators to derive valuable conclusions from the information that has been gathered.

4.2.5. Mobile device physical acquisition

Digital forensics uses mobile device physical acquisition as a backup plan in case logical extraction isn't feasible or reliable enough. The full storage capacity of a device, including both allocated and unallocated space, must be copied bit by bit throughout this operation. However, this process faces significant challenges, such as overcoming sophisticated encryption protocols, dealing with hardware-level security measures like secure enclaves, and ensuring the integrity of evidence during extraction. Additionally, the process is time-intensive and may require specialized expertise to avoid damaging the device or altering its data. When working with equipment that has robust security measures, encrypted data, or when logical procedures are unable to yield a full dataset, physical acquisition is frequently required. By using this technique, forensics analysts can get access to the device at a lower level and obtain a comprehensive picture of its storage, including any concealed or erased data that might be important for their inquiry.

On the other hand, there are risks and difficulties associated with physical acquisition. During physical collection, forensic professionals may come against substantial challenges like user passcodes and data encryption. Strong security mechanisms are built into modern devices to secure user data, and it is difficult to go around these protections. Furthermore, there's a chance that physical acquisition will change the device's status or break it in the process. Despite these difficulties specialized tools and methods are used for physical acquisition, such as the use of specialized forensic hardware made to work with a variety of mobile devices and JTAG chip-off methods, which entail accessing the device's memory through direct hardware connections (Fukami et al., 2021). These techniques are used carefully and expertly to guarantee evidence preservation and uphold the forensic procedure's integrity.

Despite the challenges and complexities in each subdomain, mobile forensics remain an essential field for uncovering crucial evidence in digital investigations. The various subdomains, from data acquisition to physical extraction, demonstrate the breadth of techniques and tools employed to navigate these obstacles. To provide a clearer perspective, the table summarizes the key techniques, their features, and the advantages and limitations within each subsection.

Thus, the table encapsulates the critical aspects of mobile forensics, offering a comparative view that highlights the strengths and challenges of each approach. By employing a combination of these techniques, investigators can adapt to the evolving technological landscape and ensure comprehensive forensic analysis. Furthermore, the integration of advanced tools and methodologies across subdomains underscores the necessity for ongoing innovation and expertise in the field, paving the way for more resilient and effective digital investigations.

4.3. Database

4.3.1. Database preservation

When it comes to preserving the integrity and admissibility of evidence in a court of law, database preservation is an essential component of digital forensics. This process, however, faces challenges such as ensuring the immutability of the database during duplication, handling large and complex databases without introducing errors, and managing encrypted databases that require decryption for analysis without altering the original content. Maintaining a database in its original format is essential for proving in court that the information is accurate and trustworthy. It is crucial to use procedures that guarantee the database is kept in perfect shape since any changes or manipulation might undermine the credibility of the evidence.

Forensics experts employ many techniques to generate duplicates of databases, comprising both symbolic and tangible backups. Physical backups duplicate the complete storage structure, including tables and indexes, whereas logical backups entail extracting data in a structured manner, such as SQL dumps. Effective database preservation requires the use of specific tools and methods. Well-known applications such as EnCase Forensic Imager, and FTK Imager are essential for making reliable forensic copies of databases (Prakash et al., 2021). With the use of these technologies, investigators may collect data without changing the original, guaranteeing that the database is retained and serving as an accurate representation of the evidence, thus preserving its credibility in court.

4.3.2. Database recovery

A key component of digital forensics is database recovery, which focuses on restoring lost or corrupted data to recreate events and find important evidence. However, challenges in this process include retrieving data from heavily damaged storage devices, dealing with incomplete or corrupted backups, and ensuring the integrity of recovered data during the restoration process. Encryption and the use of proprietary database formats further complicate recovery efforts. If investigators want to put up a thorough timeline of events, they must know how to get information from databases. A variety of data recovery methods are employed, each with a specific function in the process of recovering lost data. Sifting through transaction logs to pinpoint and retrieve certain transactions is known as log analysis; file carving is the process of obtaining data from disk space that has not been allocated. In contrast, undelete utilities are made to recover files that have been inadvertently or purposefully removed from a database.

Using certain tools and procedures, forensic professionals enable efficient database recovery. Several components of the recovery process are completed with the use of software, including Recuva, PhotoRec, and Advanced Data Recovery (Mos and MM Chowdhury, 2020; Zimba et al., 2019). For example, Recuva is excellent at recovering erased files from a variety of storage media, including databases. Because of its expertise in file carving, PhotoRec is skilled in locating and restoring deleted data by extracting file fragments. Complex recovery scenarios may be handled by a variety of software programs known as advanced data recovery tools, which provide forensic investigators the ability to recover important data from databases that have been corrupted or destroyed.

4.3.3. Analysis of database logs

One of the most important aspects of forensic investigations is the study of database logs, which is a vital tool for spotting illegal access attempts, suspicious activity, and unlawful data revisions. However, this process is often complicated by challenges such as incomplete or missing logs, inconsistencies in log formats across database systems, and the presence of vast amounts of irrelevant data that must be filtered to identify actionable insights. Additionally, attackers may attempt to tamper with logs to erase traces of their activity, requiring forensic investigators to validate the authenticity of the logs before analysis. Database logs are essential for reconstructing events and creating an activity chronology because they capture a multitude of details regarding user actions, system events, and data alterations. Forensic specialists might find patterns of conduct that can point to criminal intent or security breaches by carefully examining these logs. Understanding and identifying illegal access attempts and data alterations is essential for determining the extent and consequences of a security event, as well as for supplying important proof in court.

The forensic data analysis procedure makes use of a variety of database log formats, such as transaction, audit, and access logs. Auditing logs document system events and user activity, transaction logs document modifications made to the database. Who accessed the database when it is documented in the access logs. Every kind of log has a unique function in revealing various facets of possible security breaches. Forensic investigators use technologies like Splunk, Elasticsearch, and Kibana to do efficient log analysis (Hassan et al., 2023; Oussous and Benjelloun, 2022). Large amounts of log data may be aggregated, visualized, and analyzed with the use of these technologies, which increases the overall efficacy of digital forensic exams by assisting investigators in spotting patterns, correlations, and abnormalities that might have been ignored.

4.3.4. Analysis of database content

One of the core components of digital forensics is database content analysis, which gives investigators the tools to locate relevant information including user accounts, financial transactions, and communication logs. This process, however, is also not without challenges. Investigators must contend with encrypted or fragmented data, complex database schemas that can obscure relationships between data entities, and the risk of accessing incomplete or outdated records. Additionally, ensuring that content retrieval does not alter or corrupt the original database is critical to maintaining evidentiary integrity. To retrieve certain data that may be essential for reconstructing events and assembling evidence, this procedure entails accessing databases. During this stage, forensic professionals must comprehend the database's structure and use the right query languages to quickly explore and retrieve pertinent data.

During content analysis, a variety of database query languages and strategies are used. With the use of SQL (Structured Query Language), which is widely used to communicate with relational databases, investigators may run queries to get, modify, and examine data (Chopade and VK Pachghare, 2019). Depending on the type of database, different query languages outside SQL could be pertinent. Forensic investigators use tools like statistical analysis software to extract significant insights from massive datasets, data visualization tools to understand intricate relationships within the data, and SQL editors to create and run queries to support content analysis. When employed in conjunction, these resources enable investigators to navigate through enormous volumes of data, identify pertinent details, and unearth crucial evidence required for a thorough forensic investigation.

Thus, database forensics plays a critical role in cybersecurity investigations, offering robust methodologies for preserving, recovering, and analyzing sensitive data. Each subsection of database forensics introduces unique challenges, ranging from the complexities of ensuring data integrity during preservation to recovering lost or corrupted information and analyzing logs for suspicious activity. These challenges necessitate the use of advanced tools and techniques tailored to the intricacies of modern databases. To consolidate these insights, the table provides a comparative overview of the key techniques, tools, advantages, and limitations associated with each subdomain, offering a structured understanding of database forensics methodologies.

Hence, the above table provides a structured comparison of database forensic subdomains, encapsulating the tools, techniques, and their respective strengths and limitations. This comparative analysis highlights how forensic investigators can leverage these methods to address the nuanced challenges posed by diverse database environments. By synthesizing these approaches, practitioners can enhance their ability to ensure data integrity, recover lost information, and analyze databases effectively. This systematic understanding not only strengthens forensic methodologies but also underscores the critical role of databases in maintaining cybersecurity resilience.

4.4. IoT

4.4.1. IoT device imaging

One important component of digital forensics is IoT device imaging, which uses a variety of techniques to obtain forensic photographs for use in investigations. However, this process is often complicated by challenges such as ensuring the integrity of volatile data during extraction, dealing with proprietary device architectures, and accessing cloud-stored backups that may require compliance with privacy regulations. Additionally, physical disassembly of devices can risk evidence alteration or damage, making expert handling critical. These approaches include flash memory data extraction, JTAG imaging, and cloud backup exploration related to Internet-of-Things devices (Castro and A Perez-Pons, 2021). To preserve the integrity of the gathered evidence, the procedure necessitates giving considerable thought to device disassembly procedures. To enable a trustworthy investigative trial, proper documentation and adherence to forensic procedures are crucial during the imaging process.

IoT device imaging is a field that uses a wide range of technologies and methods. Forensics experts use devices like FTK Imager and EnCase Forensic Imager, as well as gear tailored to certain device architectures (Castro and A Perez-Pons, 2021; Prakash et al., 2021). These technologies support the thorough inspection of IoT devices in forensic investigations by assisting in the extraction and analysis of digital evidence. These techniques are constantly being shaped and improved by technological advancements, underscoring the dynamic character of digital forensics in the constantly changing Internet of Things environment.

4.4.2. IoT device artifacts analysis

Digital forensics relies heavily on the investigation of IoT device artifacts, with an emphasis on extracted data to find indications of corrupted firmware, possible data breaches, and illicit activities. This process presents unique challenges, including the diversity of IoT devices with varying architectures, proprietary firmware that may be difficult to interpret, and the presence of encrypted or obfuscated data. Additionally, analyzing communication protocols requires expertise in detecting anomalies that may indicate data tampering or unauthorized access. This entails a careful analysis of the file systems, communication protocols, and embedded system setups that are specific to Internet of Things devices. Recognizing abnormalities and possible security breaches that might compromise the devices' functioning and integrity requires an understanding of these components.

Within the field of IoT artifact analysis, forensic specialists utilize specific gadgets and methodologies to analyze and comprehend data that has been retrieved. Examining the internal operations of Internet of Things devices requires the use of tools such as IDA Pro, Binary Ninja, and firmware analysis techniques (Nadir et al., 2022). Through their insights into the codebase, these technologies aid investigators in locating malicious code, unapproved access points, and vulnerabilities. Because of the intricacy of IoT ecosystems, artifact analysis must take a sophisticated approach, needing a thorough comprehension of both software and hardware components to successfully reveal possible vulnerabilities and breaches in these networked systems.

4.4.3. IoT sensor data analytics

Determining abnormalities, reconstructing events, and tracking activity trends are all part of the process of analyzing sensor data from Internet-of-Things devices. However, this process is often complicated by challenges such as the heterogeneity of sensor data formats, the high volume of data generated by IoT ecosystems, and the difficulty of distinguishing between benign anomalies and genuine security threats. Additionally, ensuring the accuracy and synchronization of time-stamped sensor data is critical for effective timeline reconstruction and anomaly detection. By analyzing this data, investigators may gain important insights into how IoT devices are used, assembling timelines of events, and identifying anomalies that can point to security breaches or suspicious activity. Nevertheless, there are difficulties in this process, especially when handling the various sensor data types and determining their contextual importance to the IoT device's overall performance.

Many tools and methods are used to address the challenges of IoT sensor data analysis. For effective processing and interpretation of the data, Python packages for data manipulation and visualization are necessary. Furthermore, geospatial analysis tools help to comprehend the geographical context of sensor data, and statistical analysis software helps to find trends and patterns (Dorai et al., 2020; Hall et al., 2022). With the use of these analytical techniques, forensic investigators may better identify possible security risks and anomalous device behavior by extracting valuable insights from the large and diverse sensor data that IoT devices generate.

4.4.4. IoT incident response and remediation

In the context of the Internet of Things, handling security events is a multifaceted procedure that includes vulnerability discovery, device isolation, and the use of damage mitigation techniques. However, the process is often complicated by challenges such as the rapid propagation of threats across interconnected IoT devices, limited real-time monitoring capabilities in resource-constrained devices, and the difficulty of implementing scalable mitigation measures in heterogeneous IoT environments. Additionally, forensic investigations can be obstructed by incomplete logs or tampered data, further complicating response efforts. Given the interconnectedness of these devices and the inherent threats they represent to physical systems, responding to accidents involving the Internet of Things needs a sophisticated strategy. Forensic investigation to determine the extent and impact of the incident, quick threat identification and protection, and the deployment of security measures to stop it from happening again are all examples of best practices in incident response for Internet of Things settings.

In the IoT landscape, a variety of technologies and methodologies are essential to the incident and response remediation process (Itodo et al., 2021). Vulnerability scanners aid in finding configuration flaws in Internet of Things devices, while network security monitoring solutions allow for real-time surveillance and the identification of suspicious activity. Specialized IoT security platforms provide solutions for monitoring, analysis, and reaction in the event of security issues, all aimed at addressing the problems presented by networked devices. In IoT settings, these solutions help create a strong security posture by combining proactive incident response tactics with security measures that prevent possible attacks and guarantee the resilience of networked systems.

Thus, the realm of IoT forensics introduces diverse methodologies tailored to the intricate challenges posed by the rapidly evolving Internet of Things ecosystem. Each subsection of IoT forensics, from imaging devices to analyzing sensor data, presents unique opportunities and complexities that require specialized tools and approaches. These techniques ensure the reliability and integrity of digital evidence while addressing the complexities introduced by IoT's heterogeneous nature. To provide a clear overview of the current landscape, the table summarizes key tools, techniques, benefits, and limitations across the primary subdomains of IoT forensics, offering insights into their applications and impact.

The table above provides a structured comparison of the tools and techniques shaping IoT forensics. By addressing the complexities of device imaging, artifact analysis, sensor data analytics, and incident response, these methodologies empower forensic investigators to adapt to the unique challenges of the IoT domain. Leveraging these tools not only ensures comprehensive analysis but also fortifies IoT ecosystems against evolving threats. As IoT adoption continues to grow, these forensic strategies will remain integral to maintaining secure and resilient networks.

4.5. Cloud

4.5.1. Cloud architecture analysis

It is essential to have a thorough grasp of the different cloud computing models while working with cloud architecture analysis. Three separate paradigms with consequences for forensic investigations are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) (Dalal and Juneja, 2021). However, challenges in cloud architecture analysis include navigating shared responsibility models between users and Cloud Service Providers (CSPs), handling dynamic scaling and ephemeral instances that may lose critical data, and addressing the lack of standardization across CSPs. Moreover, investigators must account for multi-tenancy environments where data segregation is vital to avoid cross-contamination of evidence. To effectively traverse and recover digital evidence from cloud settings, forensic analysts must understand the nuances of these models. Furthermore, it is crucial to explore the significance of grasping the security measures put in place by Cloud Service Providers (CSPs) and data access rights (Sharmila and C Aparna, 2024). The environment for carrying out exhaustive and legally acceptable forensic examinations in cloud-based infrastructures is shaped by these factors.

Using certain tools and methods is essential to conducting an efficient examination of cloud architecture. Investigators use a range of tools, such as documents from cloud providers, to learn more about the architecture and security protocols of the cloud system they are examining. This knowledge is further enhanced by security evaluations, which offer a way to examine the risks and vulnerabilities connected to the selected cloud service. Visualization tools are essential for making complicated cloud systems understandable and helping investigators spot possible compromise sites or suspicious activity. To put it simply, a comprehensive approach to cloud architecture analysis combines an understanding of the model, awareness of CSP privacy regulations, and skillful use of investigative tools to guarantee a solid forensic study in the ever-changing cloud computing environment.

4.5.2. Cloud logging and audit analysis

One of the core concepts in the field of cloud logging and audit analysis is the understanding of the critical function that cloud logs and audit trail analysis play in detecting possible security risks. However, this process is often impeded by challenges such as the immense volume of logs generated in cloud environments, the lack of uniform logging standards across Cloud Service Providers (CSPs), and restricted access to log data due to provider policies. Additionally, log tampering or incomplete logs can complicate investigations, making the validation of log authenticity a critical step. Examining these logs is essential for identifying questionable activity, examining user behavior, and keeping an eye on resource usage in cloud settings. This process's proactive detection and mitigation of security issues, which protects the confidentiality and integrity of cloud-stored data, highlights how important it is.

A multitude of log and audit data formats are present across cloud platforms, each providing a distinct perspective on the actions taking place in the environment. Comprehending and classifying these logs are necessary preconditions for efficient analysis. Investigators use a variety of tools and approaches designed specifically for the complexity of cloud logging and audit data to make this easier. Forensic experts can't live without platforms like Splunk and LogRhythm in addition to log analysis tools tailored to individual cloud providers (González-Granadillo et al., 2021; Sheeraz et al., 2023). By enabling investigators to effectively go through enormous volumes of data, these technologies strengthen the security posture of cloud-based infrastructures by helping them identify trends, abnormalities, and possible security events.

4.5.3. Cloud Data Collection and Preservation

Knowing the best practices for obtaining and storing forensic evidence becomes essential when it comes to Cloud Data Collection and Preservation. However, this process is fraught with challenges such as accessing evidence in multi-tenant environments while ensuring no cross-contamination, obtaining necessary legal permissions across different jurisdictions, and handling encrypted data stored in cloud environments. The dynamic nature of cloud systems, where data can be relocated or deleted quickly, adds further complexity to the collection process. This entails outlining procedures for obtaining digital evidence from a variety of cloud-based sources, including databases, apps, and storage services. The complexity of cloud architecture demands specialized methods, highlighting the need to use tools and strategies suited to the difficulties faced by cloud-based data (Makura et al., 2021).

One of the most important parts of gathering and preserving cloud data is navigating the legal environment. When it comes to accessing and preserving the stored data on the cloud, forensic specialists must deal with possible difficulties and legal issues. Legality and admissibility of gathered evidence depend heavily on acquiring required permissions, adhering to service provider terms of use, and maintaining compliance with privacy standards. Investigators use a variety of techniques and tools to make this process easier, from specialist forensic platforms made for cloud settings to the legal hold features embedded into many cloud services. Furthermore, the use of data transfer technologies facilitates encrypted and easy extraction of pertinent data, which adds to a thorough and compliant approach to cloud-based forensic investigations.

4.5.4. Disk forensics imaging

Disk forensic imaging is still an essential part of cloud forensics investigations, proving its applicability even when considering cloud-based storage. However, the process faces several challenges, such as accessing disk images in shared or virtualized environments, dealing with encryption at both the disk and file levels, and handling the transient nature of virtual machines that can be deleted or modified rapidly. Additionally, ensuring the completeness and authenticity of disk images requires adherence to strict forensic procedures to prevent evidence contamination or loss. Even if cloud data storage is abstracted, forensic imaging is still necessary because it gives investigators an overview of the condition of virtual machines and cloud servers at a particular moment in time. This procedure is effective for maintaining digital evidence, enabling storage structure analysis easier, and ensuring the case's integrity.

The complexities of virtual machines and cloud servers become apparent when discussing how to obtain forensic photos in cloud systems. To ensure a thorough collection of possible evidence, specialized procedures are utilized to capture the whole disk, including both allocated and unallocated space. Tools and methods designed specifically for cloud, including imaging tools dedicated to a cloud provider, are essential. Additionally, sophisticated memory acquisition methods combined with specialized forensic applications allow investigators to navigate the complexity of cloud-based storage and extract relevant data for forensic analysis, contributing to a comprehensive and nuanced approach to disk forensic imaging in the cloud (Bhatia and J Malhotra, 2019; Wu et al., 2020).

Thus, the dynamic nature of cloud environments presents unique opportunities and challenges for forensic investigations. Each subdomain of cloud forensics, from architecture analysis to disk imaging, requires specialized approaches to navigate the complexities introduced by virtualized and multi-tenant infrastructures. Effective tools and methodologies tailored to cloud-specific challenges empower investigators to maintain the integrity of evidence, ensure compliance with legal frameworks, and derive actionable insights. The table offers a consolidated overview of the key tools, techniques, benefits, and limitations across various aspects of cloud forensics, providing a structured understanding of the current landscape.

Hence, the above table highlights the comprehensive approaches and tailored tools necessary for effective cloud forensics. By addressing the distinct challenges posed by virtualized environments, multi-tenancy, and dynamic scaling, these methodologies enhance the ability to investigate and respond to incidents in cloud settings. With the ongoing evolution of cloud technologies, forensic practices must continually adapt, ensuring investigators are equipped to safeguard data integrity, comply with legal standards, and strengthen the resilience of cloud infrastructures. This adaptive approach ensures that cloud forensics remains a cornerstone of modern digital investigations.

4.6. *Multimedia*

4.6.1. *Multimedia image forensics*

The core area of image forensics is the analysis of digital pictures to look for evidence of alteration, fraud, or manipulation. However, challenges in image forensic analysis include identifying subtle manipulations, such as deepfakes or generative alterations, and handling images with degraded quality or incomplete metadata. Moreover, distinguishing between non-malicious edits and malicious modifications often requires advanced tools and expert judgment. Images are examined using a variety of approaches, such as a review of popular altering programs and methodologies. It becomes essential in this domain to recognize the warning signals of manipulation. Image forensic analysis entails applying specialized tools and procedures intended to reveal the veracity of digital images.

Examining Exchangeable Image File Format (EXIF) data is one of those methods as it provides important details about the image's past alteration and provenance (Kumar Sahu et al., 2023). Furthermore, copy-move forgery detection algorithms are used to locate duplicate or cloned components inside an image, and error level analysis is used to find abnormalities in compression levels (Chaitra and Reddy, 2023). Together, these diverse methods help to provide a thorough evaluation of digital photos, which helps investigators make sure that visual evidence is reliable and authentic.

4.6.2. *Multimedia video forensics*

A methodical technique to examine digital recordings for indications of fabrication, editing, or manipulation is referred to as video forensic analysis (Sandhya, 2024). However, challenges in video forensic analysis include detecting advanced manipulations like frame interpolation and deepfake technology, ensuring authenticity when metadata is altered or missing, and handling large video files that require significant computational resources for analysis. Additionally, low-resolution or compressed videos can obscure critical details, complicating the detection of anomalies. This procedure necessitates a thorough comprehension of the several techniques used to determine the legitimacy of visual material. Analyzing popular video editing procedures and tools is essential for finding possible anomalies, much like picture forensics. Forensic analysts must be able to recognize clear indications of video tampering as it can provide information regarding the reliability of visual evidence.

Sophisticated tools and techniques are essential in the field of video forensic analysis to reveal the truth behind digital recordings. One such technique is video metadata analysis, which enables investigators to look at information inherent in the video file and provide details about its creation and modification history (Meshram and MK Singh, 2023). Furthermore, the detection of compression artifacts is a crucial signal as anomalies in the compression process might disclose possible modifications. By assessing the coherence and logical progression of events within a film, scene consistency analysis technologies make an additional contribution and help investigators distinguish between authentic material and edited information. These diverse methods enable forensic specialists to thoroughly examine the authenticity of digital movies, strengthening the validity of visual evidence in judicial and investigative environments.

4.6.3. *Multimedia audio forensics*

The area of audio forensic analysis involves expertise in analyzing digital audio recordings to detect any possible instances of tampering, splicing, or forgery. However, challenges in audio forensic analysis include identifying subtle manipulations, such as changes made using advanced AI tools that can seamlessly splice or alter audio content. Noise interference, low-quality recordings, and encrypted audio files can further complicate the analysis. Moreover, differentiating between intentional tampering and natural distortions caused by recording conditions often requires advanced expertise and sophisticated tools. In this regard, having a thorough grasp of techniques for analyzing audio material becomes imperative. Investigating typical audio editing tools and procedures in detail, forensic analysts look for telltale clues that might point to manipulation or tampering. Experts can uncover possible changes to audio recordings by analyzing the auditory environment, which enhances the accuracy and dependability of forensic investigations.

The methods and technologies used in digital audio recording forensics are intended to simplify the complicated nature of digital audio recordings. An essential technique for identifying abnormalities or inconsistencies is spectral analysis, which shows the frequency composition of audio signals (Pedapudi and N Vadlamani, 2023). To identify the source of voice parts in a recording, speaker identification methods are essential. Moreover, techniques for noise reduction are used to improve audio clarity, making it easier to analyze the original information accurately (Diwan and U Sonkar, 2024). Together, these techniques enable audio forensic specialists to successfully negotiate the complex world of digital audio, guaranteeing an exhaustive and trustworthy analysis of the evidence in a range of legal and investigative contexts.

4.6.4. *Social media forensics*

When exploring the broad and ever-changing world of online platforms, investigators must traverse certain problems and issues related to social media forensic investigations. Key challenges include handling the vast volume of data generated in real-time, dealing with platform-specific privacy policies and access restrictions, and ensuring the preservation of evidence before it is modified or deleted. Additionally, verifying the authenticity of social media content is complicated by the prevalence of fake accounts, bots, and deepfake media. The vast amount of data and the dynamic nature of social media make digital evidence analysis and preservation difficult. Effective multimedia forensic investigations require an understanding of the historical nature, context, and potential privacy problems related to social media information. Moreover, social networks' dynamic and linked structure contributes to their complexity, necessitating the use of sophisticated methods by investigators to track down and validate information.

The techniques used to gather and store digital evidence are vital in the field of social media forensic investigation (Arshad et al., 2022). The complexities of gathering and preserving social media material, including posts, messages, and related metadata, must be navigated by investigators. Preservation methods need to take into consideration the temporal aspect of social media data to maintain the authenticity and integrity of the multimedia evidence throughout time. Effective social media forensic investigation requires specific tools and methods. Sentiment analysis techniques help to comprehend the emotional tone and context of social media conversations, while social media archiving services allow the systematic collecting and storage of content. Link analysis tools also aid in the establishment of linkages between disparate pieces of information, which helps investigators gain a thorough grasp of the digital trail left on social media sites (Gaikwad and Dhutonde, 2023). These versatile tools and methods enable forensic specialists to successfully negotiate the intricacies of social media information, enabling a comprehensive and precise examination within the framework of legal and investigative procedures.

Thus, the domain of multimedia forensics spans diverse subfields, each with unique challenges and solutions tailored to their respective mediums. Whether analyzing digital images, videos, audio recordings, or social media content, investigators employ advanced tools and methodologies to uncover evidence of manipulation or tampering. These processes are critical for maintaining the integrity and reliability of multimedia evidence, especially in legal and investigative contexts. To provide a comprehensive overview, the table summarizes the tools, techniques, benefits, and limitations associated with multimedia forensics, highlighting the intricate yet essential processes involved in this domain.

Hence, the table encapsulates the diverse methods and technologies underpinning multimedia forensics, emphasizing their critical role in maintaining the authenticity and admissibility of digital evidence. By addressing the nuanced challenges associated with each subdomain, these tools and techniques enable investigators to uncover hidden anomalies, reconstruct events, and ensure justice in a digital era. As multimedia forensics continue to evolve alongside technological advancements, its methodologies must adapt to emerging threats, such as generative AI manipulations, to remain a cornerstone of reliable digital investigations.

5. Discussion, challenges, and future direction

Digital forensics, a crucial area of cybersecurity, faces a variety of complex issues across its several subdomains, each characterized by unique complexities and evolving techniques. From controlling device diversity, such as data extraction and retrieval across mobile, network, and cloud domains, to cracking encryption, handling dynamic data, and countering deepfakes, each area has its own unique set of obstacles as illustrated in Fig. 5. To effectively respond to security issues and acquire actionable evidence, investigators must employ sophisticated and flexible approaches that keep up with the ever-evolving digital world. Within these crucial subdomains, the next sections explore the particular

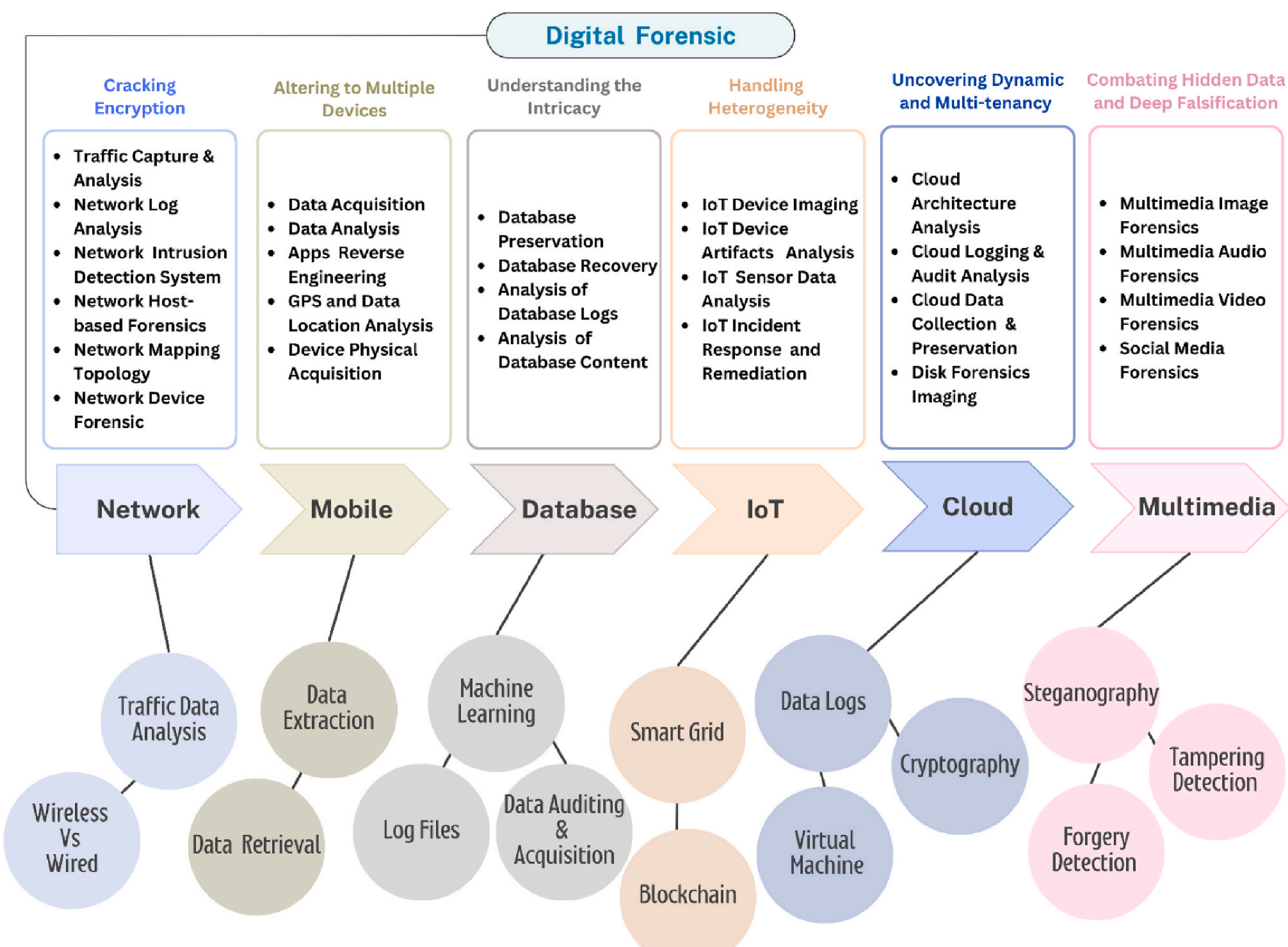

Fig. 5. Digital forensics decoded: Uncovering subdomains, unveiling techniques, and exploring future directions.

issues and prospects

5.1. Network forensics: cracking encryption

Within the field of digital forensics, network forensics focuses on the monitoring and analysis of computer network traffic to investigate security issues and collect evidence for legal purposes. Regarding "Decrypting the Encrypted," the discussion revolves around the challenges caused by the pervasive usage of encryption in communication protocols and the requirement for sophisticated decryption methods in the field of network forensics. To guarantee the secrecy and integrity of sent data, encryption is frequently used in communication protocols. Secure email communication using Transport Layer Security or Socket Layer Security (TLS/SSL), secure network connections via VPNs, and safe online surfing using HTTPS are a few examples. Although encryption is essential for protecting sensitive data, network forensics investigators have difficulties when using it. Data packet information is obscured during encrypted transmission, making it difficult to examine and evaluate network traffic for indications of criminal activity.

In network forensics, decryption methods are necessary to reveal the contents of encrypted data. This makes it possible for analysts to look at and comprehend the nature of communication, which helps to identify and reduce security problems like data breaches and cyberattacks. Adaptive approaches are needed to keep up with the exponential expansion in network data caused by connected devices, cloud services, and rising internet consumption. Wireless networks, such as Wi-Fi, often pose challenges related to signal interception and secure real-time monitoring. In contrast, wired networks prioritize secured data packet transmission and hardware integrity. Investigators must account for these distinctions to adapt forensic techniques effectively. Network forensics adaptable approaches need to take into consideration the dynamic nature of contemporary networks to meet these challenges. The enormous volumes of data found in contemporary networks must be processed, and analyzed, and useful insights must be effectively extracted from them. Network forensics becomes more challenging when infrastructure and services are moved to cloud environments. It is necessary to adjust to difficulties with data storage, network design, and access control while conducting incident investigations in dispersed and virtualized environments.

Network forensics adaptable approaches need to take into consideration the dynamic nature of contemporary networks to meet these challenges. This entails managing various encryption techniques, communication protocols, and dynamic cyber threats. The development of adaptive systems that automatically monitor network traffic, spot abnormalities, and help decode encrypted data can be aided by machine learning and artificial intelligence. The widespread use of encryption in communication protocols means that sophisticated decryption techniques are essential to network forensics. The demand for flexible approaches in virtualized and dispersed systems, the exponential expansion of network data, and the shift to cloud-based services are some of the challenges ahead. Overcoming these obstacles is critical to quickly identifying and responding to security events in the complex, encrypted environments of contemporary networks.

5.2. Mobile forensics: altering to multiple devices

Mobile devices operate on several operating systems, including iOS, Android, and Windows Mobile, and are available in a variety of shapes, sizes, and models. Forensic analysts face difficulties navigating the complexities of various devices and operating systems because of this diversity. Tools that work well for removing data from one platform might not work as well on another, which leads to the disconnected state of mobile forensics. Encryption has become widely used on mobile devices due to the growing emphasis on user privacy and security. Data is shielded from unwanted access via encryption, which makes sure that information cannot be read without the right decryption key. To overcome changing encryption techniques and carry out exhaustive investigations, forensic analysts must continually invest in cutting-edge decryption tools due to the complexity of obtaining and analyzing encrypted data.

Forensic procedures become more challenging as users depend more and more on cloud services for data storage. The physical device may not contain all the data that is important to an inquiry, therefore forensic tools may need to be modified. It becomes essential to distinguish between data that is kept locally and data that is saved on the cloud. To ensure thorough and precise analysis, forensic procedures must advance to collect data from cloud services in addition to devices. Forensic procedures need to constantly evolve due to the dynamic nature of mobile technology, which is typified by the frequent release of new gadgets, operating system upgrades, and security features. To be effective, forensic professionals and companies need to stay up to date with technology changes and modify their approaches accordingly. To improve forensic tools and stay up to date with evolving encryption standards, cloud service integration, and device-specific protection, research and development expenditures must be maintained. Keeping up with technological advancements is essential to preserving investigative effectiveness in the ever-changing field of mobile forensics.

5.3. Database forensics: understanding the intricacy

The specialty area of database forensics is devoted to looking into and examining databases to find electronic proof of security events, data breaches, or other illegal activity. The procedure entails locating, conserving, and evaluating data to precisely recreate occurrences. This task's intricacy can be due to many important variables. Because of its enormous volume and frequent changes, the dynamic and massive nature of data kept in databases presents challenges. Databases hold a variety of data, including transaction records and user information. Because data is dynamic and constantly updated or modified, it can be difficult to take a static snapshot for forensic purposes. The intricacy of databases is increased by their dynamic architecture, which can change over time due to adjustments made to tables, relationships, or data types. Comprehending these modifications is crucial for precise data interpretation.

Furthermore, databases can use several types of storage, including relational databases, NoSQL databases, and cloud-based storage, each of which calls for a unique set of forensic methods. Because cloud computing is becoming more and more popular, modern computing systems are frequently dynamic and dispersed, with databases scattered over several servers and locations. Tracing and analyzing data across the infrastructure is a specific approach required for incident investigations in such distributed environments. The utilization of data replication for load balancing and redundancy is also considered. Professionals must keep up to date on the newest encryption techniques, security precautions, and database technology, therefore adaptability in forensic approaches is essential. The efficacy of forensic approaches depends on ongoing learning, and tools need to be changed to accommodate new database architectures and security features.

Another level of complexity is introduced by the complicated access restrictions and authentication procedures that are part of databases. Databases have intricate access controls to manage user access to the data; delving into cases of illegal access or privilege escalation necessitates a thorough comprehension of these controls. The complexity is increased by authentication techniques like multi-factor authentication, passwords, and usernames. An essential component of forensic investigations is the examination of authentication records to spot any instances of illegal access. Because of the large volume and dynamic nature of data, changing database architecture, distant computing environments, the necessity for constant flexibility in techniques, and the subtleties of access restrictions and authentication systems, database forensics is therefore a challenging endeavor. To perform in-depth investigations and examine electronic evidence stored in databases, forensic professionals need to skillfully manage all these obstacles.

5.4. IoT forensics: handling heterogeneity

Digital forensics faces a distinct set of difficulties because of the diverse nature of IoT devices. IoT, which stands for Internet of Things, is a broad term that includes anything from wearables and smart household appliances to industrial sensors and linked cars. Because of this diversity, which includes variations in communication protocols, hardware, software, and data formats for forensic analysis of Internet of Things incidents is extremely difficult. It needs specific forensic techniques to address this complexity. Because of their unique features and diverse designs, IoT devices might not be easily adapted to traditional digital forensics techniques. A thorough grasp of the unique hardware configurations, data storage techniques, and communication protocols connected to each device is necessary when looking into occurrences involving Internet of Things devices.

The lack of widespread uniformity in device interfaces is one of the major challenges facing IoT forensics. IoT devices do not have a defined set of interfaces or communication protocols, in contrast to traditional computing devices. Due to the need to adjust to the unique interfaces and protocols of every IoT device, this absence makes the task of forensic investigators more difficult and adds another level of complexity to the investigation process. In addition, there are limitations with tool support. The availability and efficacy of forensic tools are impacted by the lack of defined interfaces and standards. To give researchers tools to navigate the complex terrain of IoT technologies, researchers are hard laboring creating gadgets that can manage the heterogeneity of IoT devices.

Complications arise while extracting data from IoT devices. Extracting pertinent information is a difficult operation since different devices use different data formats, storage techniques, and encryption systems. Proficiency with the subtleties of every technology is essential for forensic professionals to effectively retrieve and examine data. Finding a balance between forensic analysis and user privacy rights is essential since IoT devices frequently handle sensitive personal and corporate data. It is imperative for forensic investigators to conform to both legal and ethical norms to guarantee that their techniques respect individuals' right to privacy while facilitating the efficient investigation of incidents.

To address the difficulties caused by the diverse range of IoT devices, research is now being conducted on creating standards for data storage, communication methods, and interfaces. The objective of standardization initiatives is to facilitate forensic procedures by giving investigators a more consistent framework. These initiatives aid in the development of best practices and resources for efficient and morally sound digital investigations in the IoT age by navigating the complexity of IoT forensics.

5.5. Cloud forensics: uncovering dynamics and multi-tenancy

To precisely isolate and attribute actions, investigating events in multi-tenant cloud environments requires complex methodologies. Forensic investigations are made more difficult by multi-tenancy, which allows several users to utilize a single cloud infrastructure while maintaining their privacy. Making sure that one tenant's actions don't affect or threaten the information and operations of another is the difficult part. To preserve the integrity of the investigation, forensic methodologies must handle this by putting isolation techniques like virtualization and containerization into practice. Because cloud infrastructures are naturally dynamic, settings may be quickly changed to meet demand. In this situation, conventional forensic techniques meant for static environments might not work as well. Because cloud setups are dynamic, forensic analysts must use real-time monitoring to stay up to date with any changes. Furthermore, cloud service capabilities like versioning and snapshots have become essential for documenting the status of the environment at various times and enabling efficient forensic analysis.

In cloud systems, jurisdictional and legal difficulties are caused by the storage of data in several places and the possibility of cross-border transactions. It becomes vital to decide what region is best for an inquiry and to navigate several regulatory environments. Forensic investigators and legal specialists who are aware of the investigation's territorial implications must work closely together to overcome these hurdles. Encrypting data while it's in transit and at rest provides an additional degree of security and guarantees adherence to data privacy laws. It is essential to modify forensic techniques to take into consideration the quick changes in cloud infrastructures. It can be necessary to adapt conventional forensic techniques meant for on-premises settings to the particulars of cloud infrastructures. Tools created especially for cloud settings that can extract relevant data from cloud logs and configurations should be known to and used by cloud forensic investigators. To ensure that they can respond to and efficiently investigate cloud-related issues, investigators must get ongoing training that keeps them informed about the constantly changing landscape of cloud services and security protocols.

Thus, a thorough strategy is required when looking into occurrences in multi-tenant and dynamic cloud settings. This involves knowledge of technology, familiarity with the law, and flexibility to meet new challenges squarely. In these intricate situations, real-time monitoring, the integration of specialist technologies, and cooperation with legal experts are critical elements of an efficient cloud forensics process.

5.6. Multimedia forensics: combating hidden data and deep falsification

With the development of deepfake technology, the area of multimedia forensics examines and identifies alterations inside multimedia content and faces new difficulties. Artificial intelligence, in particular generative adversarial networks (GANs), is used by deepfakes to create synthetic media that closely resembles authentic media. It is challenging to discern between real and fake information visually or audibly due to the complexity of these algorithms. Forensic analysis requires that the metadata connected to multimedia files be kept intact. To increase the validity of their fabricated substance, deepfake creators frequently alter or remove information. As a result, preserving the dependability and correctness of metadata becomes essential to multimedia forensics, helping to demonstrate the legitimacy of the material that is being investigated.

Combating the spread of altered material requires the development of sophisticated technologies that can identify deepfakes. These tools use several methods, such as anomaly identification, pattern recognition, and examination of minute abnormalities created during the deepfake production process. Finding patterns suggestive of deepfake manipulation depends heavily on machine learning models that have been trained on large datasets of both actual and modified material. Watermarks, steganographic content, and other forms of hidden data or information may be inserted in multimedia files. To retrieve and examine this concealed data without changing the original information, forensic tools must be created. This guarantees that the integrity of the evidence is not unintentionally compromised during the investigation process.

Multimedia forensics requires constant study and improvement due to deepfake technology continuing to be changing. Effective countermeasures against deepfakes and staying ahead of future risks need cooperation between researchers, technologists, and law enforcement authorities. The legal implications of deepfakes contribute to the intricacy of multimedia forensics. In legal processes, where proving the legitimacy of multimedia information and the dependability of forensic analysis is necessary for the acceptance of evidence in courts, findings in this sector are extremely important. The continuous work in multimedia forensics helps to address and mitigate the possible damages connected with this technology, even as deepfakes continue to raise ethical and legal issues.

Moreover, Table 11 provides a detailed summary of the key challenges in each digital forensic domain, along with actionable future directions and potential technological solutions that can address these limitations. By identifying specific tools and methodologies, this table bridges the gap between current forensic challenges and emerging research areas. These technologies reflect the latest advancements in AI, blockchain, federated learning, and other innovative fields, offering domain-specific solutions for evolving cyber threats.

5.7. Application-specific future research opportunities

Despite advances in digital forensics, several domain-specific limitations continue to impede practical investigations. These unresolved challenges demand targeted, application-aware research directions to support reliable, scalable, and legally sound forensic practices.

In Network Forensics, the widespread use of encryption and evolving protocol obfuscation techniques hinder the ability to monitor malicious activities in real time. Future research should focus on developing quantum-resistant decryption methods and adaptive AI-based traffic classification models capable of interpreting encrypted or obfuscated communication flows.

Mobile Forensics faces persistent fragmentation due to heterogeneous device ecosystems and operating system-level encryption, which limit unified data acquisition. There is a need for cross-platform forensic frameworks that support encrypted data recovery and real-time cloud synchronization while respecting OS-specific security boundaries.

In Database Forensics, challenges arise from the scale and complexity of hybrid data environments, where anomaly detection and access auditing remain underdeveloped. Future efforts should prioritize AI-powered forensic engines capable of navigating unstructured datasets, as well as forensic-ready access control auditing mechanisms for distributed and dynamic database systems.

IoT Forensics continues to struggle with device heterogeneity, limited on-device storage, and the lack of standardized forensic protocols. Research should focus on lightweight, embedded forensic agents tailored for edge environments and the development of vendor-agnostic SDKs that enable consistent evidence extraction across IoT platforms.

In Cloud Forensics, multi-tenancy and data distribution across jurisdictions complicate evidence isolation and compliance. Future research must develop jurisdiction-aware extraction workflows and auto-snapshotting mechanisms that support real-time forensic readiness in SaaS and hybrid cloud infrastructures.

Finally, Multimedia Forensics faces increasing difficulty in detecting real-time manipulation, particularly in live-streamed or GAN-generated content. Future work should emphasize the creation of high-precision deepfake detection pipelines and blockchain-backed provenance tracking systems to maintain authenticity and evidentiary integrity.

Overcoming these domain-specific obstacles will also require strategic collaboration and legal foresight. Current regulations often lag behind technological capabilities, particularly in cross-border investigations. As such, innovation in AI, privacy-preserving computation, and evidentiary transparency must be matched by legal harmonization and interdisciplinary partnerships.

In summary, digital forensics operates at the intersection of technology, security, and legal accountability, requiring continuous adaptation to an increasingly complex threat landscape. As this section has demonstrated, each subdomain, including Network, Mobile, Database, IoT, Cloud, and Multimedia presents distinct technical and legal gaps. By fostering progress in explainable AI, federated forensic frameworks, and scalable, standards-aligned solutions, the field can respond to emerging threats while preserving evidentiary validity. This section offers a foundation for domain-resilient research that aligns with both operational demands and future investigative requirements.

Table 11
Challenges, research directions, and potential solutions across core digital forensic domains.

Domains	Limitations/ Challenges	Research Directions	Potential Solutions
Network Forensics	Data access is hampered by encryption.	The creation of sophisticated decryption methods.	AI-driven traffic analysis tools for encrypted packet inspection; Quantum key distribution for secure communications.
	Cloud-based traffic necessitates forensic adaptation.	Constant improvement for distributed systems.	Blockchain for secure network log integrity.
Mobile Forensics	Device encryption presents difficulties.	Purchasing cutting-edge decryption technology.	Advanced cryptanalysis techniques and AI-enhanced reverse engineering tools.
	Cloud integration challenges local data access.	Creation of instruments for smooth local and cloud analysis.	Federated cloud-mobile forensic frameworks.
Database Forensics	Efficient analysis is challenged by massive data volumes.	Creation of tools for managing big datasets.	AI-based anomaly detection in database transactions.
	Complex authentication procedures create obstacles.	Techniques for negotiating complex access restrictions.	Multi-factor authentication forensic auditing tools.
IoT Forensics	Specialized techniques are required due to heterogeneity.	Investigate standardizing forensic procedures.	Standardized forensic IoT SDKs; Privacy-preserving federated learning models.
	Limited support for diverse IoT devices.	Supporting the creation of tools and standards.	Edge computing frameworks for forensic analysis.
Cloud Forensics	Multi-tenancy complicates isolation of tenant data.	Development of sophisticated isolation tools.	Advanced containerization and virtualization forensic platforms.
	Cross-border jurisdictional issues complicate investigations.	Cooperation on global legal frameworks.	Cloud-legal jurisdiction compliance tools (e.g., GDPR-ready solutions).
Multimedia Forensics	Media legitimacy is challenged by deepfake technology.	Research into more sophisticated deepfake detection.	GAN-based detection algorithms; Blockchain for digital watermarking integrity.
	Data is hidden in multimedia files via steganography.	Tools specifically designed to find and retrieve buried data.	AI-based steganography detection and pattern recognition systems.

6. Conclusion

Despite the incomparable benefits that come with the widespread use of digital technology in our everyday lives, a dilemma involving security and privacy problems has emerged. For this, we dive into the complex field of digital forensics, closely examining six essential subdomains: mobile, network, database, IoT, cloud, and multimedia forensics. Our exploration of literature so far has demonstrated a significant lack of uniformity within these subdomains, underscoring the urgent necessity of a cohesive strategy. Due to this, our analysis uncovered the dual character of the digitization phenomena by extending its scope beyond the area of subdomains to the broader digital forensics ecosystem. One unique aspect of our investigation was the metamodeling strategy suggested to address the diversity and intricacy present in digital forensics subdomains that aimed to address the deficiency of a comprehensive perspective within the field by performing a systematic examination of inspections. Furthermore, as digitalization spreads throughout the corporate sector, novel areas like a chain of custody and financial forensics urge aggressive procedures and the issue of offering argumentation that is intelligible by humans increases with the volume of data that must be analyzed, making the integration of artificial intelligence and machine learning even more crucial. Thus, the most recent security method highlights the need for the digital forensics field to present a unified front against the constantly changing landscape of cyber threats, increasing efficacy and resilience in the future.

CRediT authorship contribution statement

Tuba Arif: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing. **David Camacho:** Writing – review & editing, Software, Formal analysis, Conceptualization. **Jong Hyuk Park:** Project administration, Investigation, Funding acquisition, Writing – review & editing, Supervision, Software, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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